Which Side Is Wrong? What Contradiction Flags Do to Retrieval-Augmented Generation Under Misinformation

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Abstract

Retrieval-augmented generation systems increasingly flag contradictions among retrieved passages to the generator. We ask what a flag must say, and how accurate its detector must be, to help against misinformation. We vary detector and flag format on fixed passages from six benchmarks, with seven generators from three families, testing hypotheses pre-registered (H1–H9 before any confirmatory data, H10–H38 in nine later addenda). A neutral flag ("these passages conflict") on gold pairs does not reliably help. Its effect depends on answer policy and model family; pooled over 34 settings, allowing for shared questions, it is -1.4 points $[-5.1, +2.3]$ (exploratory). Naming the false side helps, but on average no more than an answer key naming the wrong answer (pooled difference -0.1 points under our string metric; an outside-family judge favours the flag by $+2.5$). In a randomised test, naming the false side raises accuracy by $+17.6$ to $+33.5$ points; naming the correct side lowers it by 31.2 to 64.5. Randomised flags on true pairs help only above a break-even direction precision that rises with generator accuracy from 0.52 to 0.79. Real detectors are selective, so this threshold alone can misjudge them. Frozen before any flagged answer existed, the test's follow rates, with each cell's detections and no-flag accuracy, predict flag effects in 20 new detector–generator cells on unseen questions (mean error 1.4 points, slope 0.88). They overstate the harm on real health web pages, where naming the false side still helps and a neutral flag does not. A detector's verdict, passed as an answer key built from benchmark labels, matches its flag's mean accuracy under our metric. On genuine conflicts, direction removes valid answers, and when conflicts are rare a real detector's false flags on clean questions can outweigh its gains. We release the harness, pre-registrations and every run.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al., 2020) hands a language model passages that may disagree. Some disagreements are genuine; others are planted or stale misinformation (Zou et al., 2025; Wang et al., 2025b). Prompting a model about conflicts can help (Cattan et al., 2025). Recent systems pair contradiction detectors with flagging prompts (Wang et al., 2026) and report detection F1 and end-to-end accuracy, but not how they connect. What should the flag say, and what must the detector get right for it to help?

We fix the passages and independently vary the *detector* (heuristic, NLI, LLM, random placebo, oracle, and synthetic detectors of chosen quality) and the *format* passing its output to the generator: a neutral "these passages may conflict" flag; a flag saying one side is false but not which; a *directional* flag naming the likely misinformation; or removal of the suspect passage. After pilots on RAMDocs (Wang et al., 2025b), we pre-registered H1–H9 and tested them on 300 held-out questions. Nine addenda (H10–H38), each committed before its runs, added ConflictBank (Su et al., 2024) (including a majority-misinformation variant); RAGuard (Zeng et al., 2025), whose misleading passages are real Reddit posts; real health web pages (HealthContradict); and seven generators from OpenAI, Anthropic and Qwen. Controls separate what a flag conveys: forced choice, an answer key, randomised direction, the detector's verdict as an answer key and its model's own answer as a hint. A further 30 cells cross six detectors with six generators. WikiContradict (Hou et al., 2024), whose conflicts are genuine, is the contrast case.

Our findings:

- **A correct neutral flag does not reliably help.** Asked for every valid answer, OpenAI and Qwen models also give the misinformation; asked for one answer, some generators abstain

on misinformed and correct answers alike; under forced choice the effect is small. Claude models move less; the minimal prompt is most harmful.

- **An oracle's direction is a ceiling.** Naming the false side helps on every benchmark that labels one, but on average it is close to an answer key naming the wrong answer (pooled over 9 comparisons, which vary individually; equivalent within ± 3 under our string metric, not under an outside-family judge).
- **A directional flag pays only above a break-even direction precision.** When direction is randomised (H31), a wrong flag costs more than a right one gains, because generators follow both at similar rates and an accurate generator has more correct answers to lose. The break-even direction precision, which calibrates flags on true pairs, rises from 0.52 for gpt-4o-mini to 0.79 for gpt-6.1-terra. For real detectors, precision alone can mislead. These follow rates and each detector's flags, frozen in advance, predict the flag's effect on unseen questions (H32), though not on HealthContradict.
- **Direction is unsafe on genuine conflicts and costly when conflicts are rare.** On WikiContradict the LLM directional flag and removal cut the share of answers giving both sides (H28). At a 5% share of conflict questions, the LLM detector's flag has a negative estimated effect in 22 of 31 settings (exploratory).

On CONFLICTS, the one benchmark of real search results, no condition changes accuracy significantly. We release the harness, pre-registrations and every run.[1]

2 Related Work

Knowledge conflicts. Models can prefer memorised knowledge over contradicting context (Longpre et al., 2021), yet coherent contradicting evidence readily persuades them (Xie et al., 2024), depending on their prior's strength (Wu et al., 2024). Huang et al. (2025) train models to decide whether to trust context or memory, including real incorrect Reddit contexts. Context-faithful prompting changes how readily models follow conflicting context or abstain (Zhou et al., 2023). Among the conflict types of Xu et al. (2024), we study inter-context conflicts, as do Chen et al. (2022), who find models tend to rely on a single passage when evidence conflicts, and Wan et al. (2024), who find models weigh relevance over credibility cues.

Benchmarks. RAMDocs (Wang et al., 2025b), built on AmbigDocs (Lee et al., 2024), mixes genuine ambiguity, misinformation and noise. CONFLICTS (Cattan et al., 2025) annotates search results with conflict types and each type's desired behaviour. WikiContradict (Hou et al., 2024) pairs genuinely contradicting Wikipedia passages. FaithEval (Ming et al., 2025), ConflictBank (Su et al., 2024), ConfRAG (Yuan et al., 2026), QACC (Liu et al., 2025) and WhoQA (Pham et al., 2024) cover related settings. RAMDocs and ConflictBank label which side is misinformation; WikiContradict does not.

Telling the model about conflicts. Prompting a model to reason about the conflict type helps, and supplying the gold type helps much more (Cattan et al., 2025). On WikiContradict, models acknowledge both sides more often when told the passages may conflict (Hou et al., 2024). Ragability (Gross et al., 2026) finds such a hint helps on contradictory contexts but hurts on *consistent* ones, where it is false. With our neutral flag the hint is true and the input contradictory, yet accuracy falls: the hint does not say which side to trust. RGB (Chen et al., 2024) warns models that passages may contain errors, yet they rarely detect them; our neutral flag is that warning, made pair-specific. Pan et al. (2023b) test detection and "vigilant" prompting against LLM-generated misinformation, and Pan et al. (2023a) call for misinformation-aware QA. Source metadata and credibility signals also shape conflict resolution (Chiang and Lee, 2024; Ge et al., 2025).

Telling the model which passage is wrong. Closest to our directional format, Hong et al. (2024) pass a discriminator's marks on perturbed passages into the generator's prompt, making the generator more robust to counterfactual noise. CrAM (Deng et al., 2025) rescales attention by passage credibility, RA-RAG (Hwang et al., 2025) estimates source reliability, and Weller et al. (2024) defend open-domain QA against disinformation. TACS (Yu et al., 2024) detects untruthful context and masks it, close to our removal. FaithfulRAG (Zhang et al., 2025) models fact-level conflicts before generating. These works, with Pan

[1] Code, the pre-registration and all run outputs are in the anonymised supplementary material and will be released publicly on acceptance.

et al. (2023b)'s finding that vigilant prompts give limited protection, already show a reliability signal can help more than a warning; some vary the signal's quality (Hong et al., 2024; Pan et al., 2024). Models also follow injected hints and asserted views, wrong ones included (Turpin et al., 2023; Sharma et al., 2024). Advice improves a decision-maker only when its errors fall where the decision-maker is less accurate (Bansal et al., 2021; Mozannar and Sontag, 2020), the condition our break-even makes quantitative. Whether abstaining is a failure depends on a wrong answer's cost (Wen et al., 2025; Song et al., 2025). We add a randomised direction manipulation on the same pairs, answer-key controls that separate the verdict from the marking, its cost on clean questions, and a break-even direction precision per generator, showing why it can misjudge selective detectors.

Detection, resolution and poisoning. ConflictRAG (Wang et al., 2026) trains a pairwise detector with an LLM fallback and reports pair-level F1 alongside answer gains; its NLI and chain-of-thought baselines resemble two of our detectors. LLMs detect conflicts but struggle to pinpoint them (Gokul et al., 2025; Wang et al., 2024; Jiayang et al., 2024). They favour the more frequent side, the side matching their prior, and LLM-generated passages (Jin et al., 2024; Kortukov et al., 2024; Tan et al., 2024), and are distracted by irrelevant context (Shi et al., 2023; Schuster et al., 2026). TCR (Ye et al., 2026), context-aware decoding (Shi et al., 2024), EvidentialRAG (Hossain et al., 2026) and credibility-aware generation (Pan et al., 2024) learn or decode around unreliable passages. MADAM-RAG (Wang et al., 2025b), Astute RAG (Wang et al., 2025a), Self-RAG (Asai et al., 2024), Corrective RAG (Yan et al., 2024) and InstructRAG (Wei et al., 2025) handle conflicts without an explicit detector. A few injected passages can steer a RAG system (Zou et al., 2025); RobustRAG (Xiang et al., 2024) isolates passages for certifiable robustness, and TrustRAG (Zhou et al., 2025) filters suspected poisoned ones, the analogue of our removal. We ask instead how to pass a detector's output to the generator.

3 Framework

A question q comes with retrieved passages $P = (p_0, \ldots, p_{n-1})$. A *detector* returns flagged pairs $F \subseteq \{(i, j) : i < j\}$; a *directional* detector also returns, for each flagged pair, a *suspect* set $S_{ij} \subseteq \{i, j\}$, the side it believes is misinformation. A *format* turns (P, F, S) into the generator's prompt. We vary detectors and formats independently. Every prompt shares a skeleton: passages, question and an instruction to end with a `Final answer(s):` line. The *default* skeleton asks for every valid answer when the passages support several (RAMDocs contains same-name ambiguity, so a correct answer may need several entities). *Minimal* drops that policy; its answer line allows several answers without inviting them. *Single* says the question has exactly one correct answer (true of ConflictBank by construction). All three allow "unknown" when no passage answers the question; the *forced* skeleton is *single* without that option (Appendix A).

3.1 Formats

No flags (plain). Passages and question only.

Instruction. Plain plus a generic, detector-free instruction to check the passages for contradictions: v1 ends "if a disagreement reflects genuinely different valid answers, report each one"; v2 asks the model to judge which passages are reliable and omit answers supported only by unreliable ones.

Flag (neutral). Each flagged passage gets the header `[POTENTIAL CONFLICT with passage(s): j]`, followed by a list `Passages i and j: possible contradiction`. The format is CAAG's original; every detector's flags render identically, so detectors differ only in *which* pairs they flag.

One-wrong flag. As Flag, but adds that exactly one passage of the pair contains misinformation, without saying which.

Directional flag. As Flag, but suspect passages are headed `[LIKELY MISINFORMATION; conflicts with passage(s): j]`, and the list says which passage likely contains misinformation.

Controls. *Pick*: the neutral flag plus "decide which side is better supported and answer with that side only". *No pairs*: flagged passages are marked, not which conflicts with which. *Answer key*: plain plus "The answer X is false" for each wrong answer, with no passage marked.

Remove (suspect). Each flagged pair's suspect passages are deleted; for a pair without a suspect, its second passage, as the original CAAG REMOVE strategy does without retrieval scores.

Single call. No detector: one call writes `Suspect passages: . . .`, then answers from the other passages, at half the calls of a detector pipeline.

Closed book. No passages: what the generator knows without retrieval.

3.2 Detectors

CAAG (Contradiction-Aware Attention Gating) is a heuristic detector from an earlier, unpublished RAG system, code unchanged. CAAG scores similar pairs by negation, entity-mismatch and temporal cues, runs locally and gives no direction (Appendix B).

NLI is `nli-deberta-v3-base` (He et al., 2023), a cross-encoder scored in both orders; a pair is flagged if the larger contradiction probability exceeds 0.5. It gives no direction.

LLM is a single call that lists contradicting pairs as JSON, distinguishing them from ambiguity, complementary and temporal relations. It names the likelier misinformation side (or "both" or "unclear"). Its model is gpt-4o-mini (temperature 0) unless stated; in the crossed design gpt-4.1, gpt-6-luna, gpt-6.1-terra, Claude Haiku 5.5 and Claude Sonnet 5.5 run the same prompt.

Random is a placebo flagging each pair independently at CAAG's per-pair flag rate.

Oracle flags exactly the gold contradiction pairs (as the benchmark's labels define them) and names the gold misinformation side. The *inverted oracle* (adversarial control) flags the same pairs but names the correct side.

3.3 Synthetic detectors

To separate a detector's accuracy from its identity, we build gold-label detectors with a chosen pair-level precision $p \in \{0.25, 0.5, 0.75, 1\}$, recall $r \in \{0.25, 0.5, 1\}$ and direction accuracy $d \in \{0.5, 0.75, 1\}$ (36 grid cells), each rendered as a directional flag and as removal. All cells reuse the same random draws, so they nest and cross-cell comparisons are paired (Appendix G).

4 Experimental Setup

Data. **RAMDocs** (Wang et al., 2025b) has 500 questions, each with 1–12 documents (5.5 on average) typed *correct* (supports a gold answer, possibly for a different same-name entity), *misinfo* (supports a wrong answer) or *noise*. A pair is a *contradiction* if one side is misinformation and the answers differ, *ambiguity* if both are correct with different answers, *consistent* if the answers match, and *unrelated* if either is noise. The rule does not align entities: at least 37% of the 709 hold-out oracle pairs likely join passages about different entities; on the 41 single-entity questions the oracle neutral flag still costs 4.4 points (Table 41). A *conflict* question has a contradiction pair; a *clean* one has none. Some documents typed *correct* contain a wrong answer; questions whose only sign of conflict is such a document count only in "All" (Appendix F).

Benchmark	What it supports	What it cannot support
RAMDocs	neutral and directional flags (H1–H12); confirmatory split	natural misinformation; entity-aligned pairs
ConflictBank	answer policy; break-even (H31); crossed cells (H24, H30); prospective test (H32)	real documents
RAGuard	what generators do with transferred labels	factual-oracle conclusions
Health-Contradict	flag formats on real pages (H37)	verified labels
WikiContradict	coverage of competing sources (H28)	which answer is factually wrong
CONFLICTS	null results only	any positive claim

Table 1: The evidence hierarchy. The break-even direction precision rests on ConflictBank alone; the other benchmarks add breadth for the flag-format results, not independent confirmations.

ConflictBank (Su et al., 2024): for a Wikidata fact, an LLM writes one passage for the true object and one for a substituted object. We sample 400 questions round-robin over relations: 300 conflict questions (correct passage, misinformation passage, two distractors from other subjects) and 100 clean ones (correct passage, three distractors). Every question has one gold answer. A *majority* variant gives each conflict question the correct passage and ConflictBank's three wrong-object passages. ConflictBank's misinformation, temporal and semantic conflicts all support the substituted object and count as misinformation. ConflictBank had no pilot: its registered tests come from the addenda (Table 3), and everything else on it is exploratory.

RAGuard (Zeng et al., 2025) pairs PolitiFact claims (true or false) with Reddit posts that an LLM labelled supporting, misleading or unrelated, so the misleading side is real online text. We keep 294 claims with at least one supporting and one misleading post (up to two of each, plus two unrelated posts) and 100 with no misleading post; the question is whether the claim is true. Its post labels are behavioural, set by whether GPT-4 judged

the claim correctly with the post in context (Zeng et al., 2025, §3.3). They do not verify that two posts contradict or that the misleading one is false, so its "oracle" is a label transfer, not a factual oracle (audit in Appendix D).

HealthContradict (Zhang et al., 2026) pairs a yes/no health question with one real web page per side; the page agreeing with the topic's evidence-based answer is typed correct. We use 247 items over 81 topics, pages cut to 4,000 characters, labels as released (Appendix E).

WikiContradict (Hou et al., 2024) has 253 questions, each with two genuinely contradicting Wikipedia passages; the reference answer is *both* sides. The original evaluation also asks that an answer acknowledge the incompatibility and favour no side without support (Hou et al., 2024, §3.4), so we measure coverage of the competing sources (original rubric in Appendix D). We run 237 automatically scorable **CONFLICTS** (Cattan et al., 2025) questions (Appendix F).

Pre-registration and splits. We developed the method on a random 200-question RAMDocs pilot split. Before running anything on the remaining 300 questions (139 conflict, 126 clean, 35 excluded), we committed a pre-registration fixing the conditions, metrics, corrections and the confirmatory hypotheses, H1–H9. Main-text RAMDocs results all use this split; pilot results appear only in the appendix. Later addenda registered H10–H31, each before its runs but after the confirmatory results were known. The last addendum (§19) registered H32–H38: a prospective test on 400 unseen ConflictBank questions (predictions committed before any flagged answer existed), a full judge re-score and HealthContradict. Re-analyses of seen data, some unregistered (pooled key comparisons, break-even, the crossed-cell account, prevalence, FDR), are exploratory (Table 3), as are unregistered comparisons, including every real-detector row of Table 6. The repository's commit history is the only timestamp. Appendix F lists every deviation.

Generators and baselines. The main generator is `gpt-4o-mini` (OpenAI et al., 2024) (temperature 0.1, at most 256 output tokens, seeds 0–2). We replicate key conditions with `gpt-4.1`; Qwen2.5-7B-Instruct (Qwen Team et al., 2024) (4-bit, local); two reasoning models, `gpt-6-luna` (3 seeds) and `gpt-6.1-terra` (seed 0; one sample); and Claude Haiku 5.5 (3 seeds) and Claude Sonnet 5.5 (seed 0). The reasoning and Claude models run at their default temperature with up to 4,096 output tokens. On ConflictBank the primary prompt is the single skeleton; the default and minimal skeletons are robustness checks. We rerun the oracle and LLM-detector conditions with the minimal skeleton. The LLM detector is gpt-4o-mini except in the crossed design (H24), where six detectors and six generators form 30 of the possible pairs on ConflictBank; we also run it on Qwen2.5-7B with gpt-4o-mini generating. Two controls test whether a crossed cell's gain is only a stronger model's answer (H27): an own-answer hint (the detector model's plain answer) and the detector model as the generator. An artefact probe (H29) reruns two detectors on ConflictBank with passage order reversed and with every passage paraphrased by one model. MADAM-RAG (Wang et al., 2025b) is ported unchanged (three debate rounds). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) use their published prompts and defaults; since both return one answer, we also run *multi* variants asking for every valid answer (Appendix F). A detector pipeline counts as one detector call plus one generator call.

Metrics. On RAMDocs an answer is correct if it includes every gold answer and no wrong answer (Wang et al., 2025b). Our primary metric reads "includes" literally: after normalisation, each gold or wrong answer must be a whole-word span of the extracted final answer. A bidirectional variant, which also counts a predicted answer contained in a gold one, is the robustness metric: a result is *robust* if it also holds under it. We chose the one-way metric before the confirmatory run because it is closer to MADAM-RAG's published score (Appendix F), and we compare conditions only within our harness. On WikiContradict an answer is correct if it contains both sides (bidirectional matching); on ConflictBank and RAGuard we use the RAMDocs metrics. We also report misinformation share (answers containing the misinformation), abstention share ("unknown") and selective accuracy (correct over answered). An LLM judge (gpt-6.1-terra) agrees with the primary metric on 92.6% of 162 stratified conflict-question outputs ($\kappa = 0.85$); the metric is slightly conservative, missing some paraphrases. We release an unlabelled 200-output human-audit sheet.

Statistics. We average each question's score over seeds and report paired differences from the

plain format, with bootstrap 95% CIs over questions (10,000 resamples) and two-sided sign-flip permutation tests (Efron and Tibshirani, 1993; Dror et al., 2018) (10,000 permutations, extended to 200,000 when none reaches the observed statistic). A registered test takes the Holm-adjusted p (Holm, 1979) of its *family*, the tests one hypothesis registers (e.g. H15 across generators). The 636 exploratory comparisons of the condition tables form one family, and the hypothesis tables add a Benjamini–Hochberg q over all registered tests. Only results significant under Holm are called significant or set in bold; CIs are unadjusted and not read as significance. A null difference is not read as equivalence without a TOST (Lakens, 2017) at ± 3 points. Selective accuracy is a ratio, tested by swapping conditions within questions. Pooled estimates use a multilevel model with random effects for generator, benchmark and setting. Appendix F gives the minimum detectable effect.

Cost. Token counts are from the API's usage reports. All experiments, pilots included, cost $204.18 in API fees (prices checked on 2026-10-27; Appendix F).

5 Results

Tables 5 and 6 give every condition and the hypothesis tables every registered test. Table 3 marks each claim as registered before any confirmatory data (H1–H9), in an addendum before its runs (H10–H38), or a re-analysis (Appendix B); Table 1 says what each benchmark can support.

5.1 What a neutral flag does depends on the answer policy

A neutral flag on exactly the gold contradictions lowers gpt-4o-mini's RAMDocs conflict-question accuracy by 6.7 points [−10.8, −3.1] from 22.1% (H2), and gpt-4.1's by 8.6 (H9; split by answer policy in Table 2).

Asked for every valid answer, the model adds the misinformation. Under RAMDocs' default prompt, the flag raises gpt-4o-mini's misinformation share from 47.7% to 61.9%: the model reports both sides. Every OpenAI and Qwen generator moves the same way; both Claude models the other way (Haiku on RAMDocs 55.2% to 47.2%).

Asked for one answer, it abstains. Under the one-answer prompt, the flag costs gpt-4o-mini 18.7 points on ConflictBank, gpt-4.1 8.0 and gpt-6-luna 9.4 (H15, H19), and does not significantly change accuracy for gpt-6.1-terra, Qwen2.5-7B, either Claude model (H25) and every generator on RAGuard (H26). Of gpt-4o-mini's misinformed answers, 45% abstain with the flag, as do 37% of its correct ones (H21). Under forced choice (H20) its effect is significant only for Claude Haiku (+3.9).

Across settings. In a multilevel model over the 34 settings (Table 38; exploratory) with their sampling covariance (shared questions), the pooled effect is −1.4 points [−5.1, +2.3] ([−2.2, −0.6] conditional on these settings). By prompt, it is −12.1 [−23.2, −1.0] (minimal), −3.5 (multi-answer), −0.4 (one-answer) and +1.5 (forced choice); Claude shifts it up (+4.6; codings chosen after seeing data; utility in Table 39). On WikiContradict, where giving both sides is correct, the flag raises accuracy (H8: +6.5).

5.2 Naming the false side: what an oracle can deliver

With the same oracle pairs, the directional flag beats the neutral and one-wrong flags in every registered comparison but one (H3, H9–H12, H15, H16, H19, H25, H26; gpt-6.1-terra on RAGuard is not significant). On ConflictBank's one-answer prompt it gains 17.0 to 36.4 points for the OpenAI generators and 16.3 to 23.3 for Claude. It helps under forced choice (H20) or majority misinformation, and on RAGuard (H26). On RAMDocs it gains 13.4 points with gpt-4o-mini and up to 39.6 with gpt-6.1-terra, but not significantly for Qwen2.5-7B or gpt-4o-mini under the minimal prompt (H10). On HealthContradict's real web pages it gains +5.5 to +13.8 points (H37a); the neutral flag, nothing significant (H37b; Table 44). The one-wrong flag helps only under forced choice (H16, H20).

Oracle rows are a ceiling, not a detector result. An answer key ("the answer X is false", no passage marked) does as well: no registered comparison differs significantly (H23; −5.0 to +5.3 points). Pooled over the 9 comparisons (each benchmark's questions bootstrapped jointly), the flag's advantage is −0.1 points [−1.5, +1.2], equivalent within ±3 on average (90% CI [−1.3, +1.0]; exploratory). Equivalence rests on the matcher: an outside-family judge re-scoring all 23,204 pooled-test responses puts the flag ahead

	No flags				Oracle, neutral flag				Oracle, directional flag			
Generator	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.
RAMDocs, multi-answer prompt												
gpt-4o-mini	22.1	47.7	3.4	22.8	15.3	**61.9**	5.0	16.2	35.5	**14.9**	2.9	**36.5**
gpt-6-luna	14.4	72.2	1.2	14.6	11.3	76.7	0.7	11.4	53.5	**11.3**	1.4	**54.3**
Claude Haiku 5.5	25.2	55.2	2.2	25.7	26.4	**47.2**	2.2	27.0	53.5	**6.7**	2.9	**55.1**
gpt-6.1-terra	10.8	75.5	2.2	11.0	10.1	81.3	1.4	10.2	50.4	**13.7**	2.2	**51.5**
Claude Sonnet 5.5	45.3	24.5	2.2	46.3	45.3	21.6	2.2	46.3	59.7	**3.6**	1.4	**60.6**
ConflictBank, multi-answer prompt												
gpt-4o-mini	21.6	72.8	0.3	21.6	12.8	**83.2**	0.9	**12.9**	49.4	**41.2**	1.2	**50.1**
gpt-4.1	13.0	82.7	0.7	13.1	6.7	**90.7**	0.3	**6.7**	36.0	**58.3**	1.0	**36.4**
Qwen2.5-7B	23.7	67.0	6.0	25.2	15.3	**79.0**	5.0	**16.1**	41.3	**52.0**	13.0	**47.5**
ConflictBank, one-answer prompt												
gpt-4o-mini	51.1	31.0	6.7	54.8	32.4	**17.7**	**41.0**	55.0	81.2	**4.2**	**2.1**	**83.0**
gpt-6-luna	53.6	24.3	11.9	60.8	44.1	**17.4**	**28.9**	62.0	90.0	**0.0**	**0.8**	**90.7**
Claude Haiku 5.5	66.8	22.0	1.2	67.6	67.4	22.1	0.9	68.0	90.1	**0.2**	1.1	**91.1**
gpt-6.1-terra	73.0	18.0	1.7	74.2	70.7	15.7	4.7	74.1	90.0	**0.7**	0.3	**90.3**
Claude Sonnet 5.5	75.7	15.7	1.0	76.4	78.7	12.0	1.7	80.0	92.0	1.0	1.7	93.6
gpt-4.1	64.7	27.3	1.7	65.8	56.7	21.7	14.0	65.9	91.3	0.3	1.7	92.9
Qwen2.5-7B	50.3	35.3	5.0	53.0	47.0	28.7	15.0	55.3	75.7	13.0	1.3	76.7
ConflictBank, forced choice												
gpt-4o-mini	55.4	31.6	0.1	55.5	55.2	31.9	0.4	55.5	79.7	**6.7**	0.0	**79.7**
gpt-6-luna	64.6	25.8	0.0	64.6	66.1	22.0	0.0	66.1	91.0	**0.1**	0.0	**91.0**
Claude Haiku 5.5	69.3	20.9	0.0	69.3	73.2	**16.8**	0.0	**73.2**	90.4	**0.0**	0.0	**90.4**
gpt-6.1-terra	77.7	14.7	0.0	77.7	77.3	13.3	0.0	77.3	92.3	**0.3**	0.0	**92.3**

Table 2: Risk and coverage on conflict questions (%): strictly correct answers, answers that contain the misinformation, abstentions ("unknown"), and selective accuracy (correct answers over non-abstained answers, pooled over questions and seeds). **Bold**: the change from no flags is significant (H21 for ConflictBank one-answer and forced; the paper's exploratory family elsewhere). The answer policy decides what a neutral flag does: under a prompt that asks for every valid answer it adds the misinformation to the answer; under a one-answer prompt that allows "unknown" it moves answers, misinformed and correct, into abstention; under forced choice it has no abstention to trade.

by $+2.5$ [$+0.9$, $+4.2$], not within ± 3 (H38b). Individually, equivalence holds in only 2 of 9 comparisons (between-comparison τ 2.1 points, REML); the arms disagree on 11.7% of question–seed pairs, two no-flag seeds on 6.7%.

5.3 With a real detector, its wrong flags decide

Registered results differ by generator. With the gpt-4o-mini LLM detector, the directional flag significantly raises accuracy over all ConflictBank questions for 1 of 7 generators (gpt-6-luna, $+6.1$) and significantly lowers it for 1 (gpt-6.1-terra, -8.8; H15, H19, H25). It helps all 4 generators under majority misinformation, and on RAGuard both Claude models but no OpenAI one (H26).

A wrong flag costs more than a right one gains. In a randomised test (H31), each ConflictBank conflict question was flagged on its true contradiction, naming the false side for a random 159 the correct side for the other 141. A right flag raises accuracy by $+17.6$ to $+33.5$ points and a wrong one lowers it by 31.2 to 64.5 (Figure 1, left). The net effect is significantly negative for 3 of 4 generators at an even mix (H31c). For right- and wrong-flag effects R and W, randomised flags on true pairs do net harm below a break-even direction precision $p^\star = -W/(R - W)$: 0.52 [0.43, 0.60] for gpt-4o-mini, 0.57 for gpt-6-luna, 0.67 for Claude Haiku and 0.79 [0.72, 0.85] for gpt-6.1-terra (Table 25; exploratory). Follow rates overlap except gpt-6.1-terra's, so $p^\star$ rises mainly because an accurate generator has more to lose. The threshold is a calibration, valid for a real detector only if it flags typical questions and nothing else.

Real detectors are selective. Across H24's 30 cells (Table 42), comparing precision with $p^\star$ can mislead. The gpt-4o-mini detector's precision (69%) is just above Claude Haiku's break-even, implying $+1.3$ points, but the cell's effect is -4.1: its flags on a correct passage (30% of questions) change accuracy there by -41.4. The follow-rate account $\hat{\Delta} = \frac{1}{n}\sum_i [d_i(1 - a_i) f_R - w_i a_i f_W]$ (d_i: the flag names the false side; w_i: it marks a correct

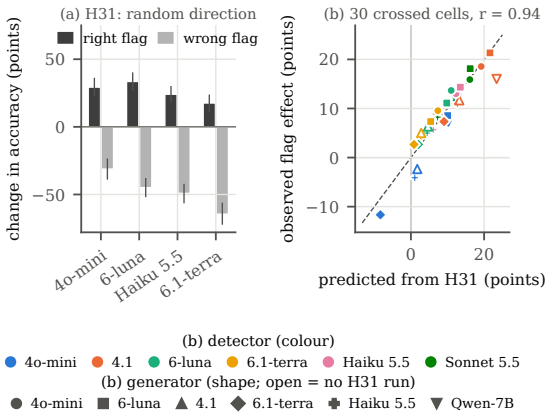


Figure 1: Left: randomised direction (H31; ConflictBank, one-answer prompt): change in accuracy (95% CIs) on conflict questions given a right flag (naming the false side) or a wrong one (the correct side). Right: predicted (from H31's follow rates) against observed effect in the 30 crossed detector–generator cells (filled: own rates; open: mean rates, no H31 run); dashed: $y = x$. Exploratory and retrospective: the account was defined after seeing the cells (Table 36).

passage; a_i: no-flag accuracy; f_R, f_W: H31's follow rates), defined after the cells were seen, tracks the 22 cells with an H31 generator (mean error 1.5 points; Figure 1). The registered H24 correlation is uninformative, since generator accuracy enters both sides. The least precise detector (gpt-4o-mini, 68.8%) lowers accuracy for the four most accurate generators, significantly for gpt-6.1-terra (H19).

Frozen in advance, the account predicts new cells (H32). On 400 unseen ConflictBank questions, five detectors (one new) crossed four generators, with effects predicted from detections and no-flag answers before any flagged answer existed. Over the 20 cells the slope is 0.88 [0.77, 1.00] and the mean error 1.4 points, against 4.1 for generator means (cell-level rates alone: 1.8). The sign is right in all 18 cells predicted at $\geq$3 points, but only 1 of 2 predicted harms is significant (H33 fails). The account overstates harm on clean questions (-4.9 vs -1.4) and on HealthContradict (gpt-6-luna -19.4 vs -8.6; Table 43, Appendix E).

In mean accuracy, the detector's verdict carries most of its contribution. Passed as an answer key, the verdict matches the directional flag (H30, no cell significant): pooled over the 22 cells the flag is ahead by $+1.2$ [-0.1, $+2.5$], equivalent within $\pm$3 under the matcher but not under the judge ($+2.4$ [$+1.1$, $+3.7$]; H38a). The key's X is the answer the detector's suspect passage supports, by the benchmark's labels: an information-content control, not a deployable format. The flag beats the detector model's own-answer hint by $+5.3$ [$+3.2$, $+7.6$] (H27).

With rare conflicts, false flags can outweigh gains. The LLM detector flags clean questions (76.2% of RAMDocs'). At a 5% share of conflict questions its flag's estimate is negative in 22 of 31 correlated settings (significantly in 11; Table 26).

Genuine conflicts and baselines. Reordering (Liu et al., 2024) or paraphrasing passages does not significantly change the detectors' hit rate (H29). Where both sides are valid, direction does damage (H28): on WikiContradict the LLM directional flag lowers the share of answers giving both sides by 6.5 to 29.0 points (and of answers correct under the benchmark's own rubric by 6.7 to 31.6; Appendix D). On RAMDocs' same-name questions, removal costs every generator 21.2 to 29.3 points. Baselines: Appendix B.

6 Discussion

Against misinformation, a neutral flag leaves the model to its answer policy, so in our settings it suits a one-answer policy with a measured cost for wrong answers (Wen et al., 2025). A directional flag is advice generators follow, right or wrong, as with sycophancy (Sharma et al., 2024). Such advice pays only above a break-even direction precision, as in human–AI complementarity (Bansal et al., 2021; Mozannar and Sontag, 2020). That threshold calibrates randomised flags on true pairs. For selective detectors, the frozen follow-rate account predicted unseen cells but not real web pages.

Limitations

Models and detectors. We test seven generators from three families: four OpenAI models (gpt-4o-mini, gpt-4.1 and two current reasoning models), Claude Haiku 5.5 and Sonnet 5.5, and a 4-bit Qwen2.5-7B. Gemini serves only as a judge, and we run no large open model. Qwen2.5-7B, gpt-6.1-terra and Claude Sonnet ran with one seed, as did gpt-4.1 outside RAMDocs. The reasoning and Claude models ran only at default temperature, sampling nondeterministically. Claude ran through Anthropic's OpenAI-compatible endpoint. The LLM detector is a single prompt. The published baselines are reimplemented (Astute RAG has no official code; restoring TrustRAG's system message changed little), NLI sees only a sentence-sized window, and we compare against no trained detector such as ConflictRAG's (Wang et al., 2026). The crossed reliability test (H24) covers 30 detector–generator cells on one benchmark with six detectors, so detector-level tests have little power (an exact permutation has 720 arrangements). The generator's own accuracy dominated the registered H24 correlation. Concurrent runs gave a detector slightly different outputs across cells (identical for at least 90% of questions); restricting each cell to those moves no cell's gain by more than 1.3 points. The follow-rate account was defined after seeing the crossed cells. Its prospective test (H32) uses unseen questions but the same benchmark, prompt and generators. The account overstates harm on clean questions and on HealthContradict. The break-even direction precision is a calibration for randomised flags on true pairs, not a validated rule for selective detectors.

Data and power. Confirmatory evidence comes from two benchmarks: RAMDocs (300 questions, 139 with conflicts) and ConflictBank (400 sampled, 300 with conflicts). In both, misinformation is synthetic, made by editing or generating text around a substituted answer; ConflictBank's passages are LLM-written on both sides. The current reasoning models distrust ConflictBank's passages when judging them (Appendix C), which suggests generated benchmarks may not measure how such models treat real documents. Reversing and paraphrasing passages do not significantly change the detectors' hit rate (H29), but equivalence within ± 5 points is shown for only one of four changes, so position, surface style and deeper regularities of generated text remain possible sources of the detectors' direction. The majority variant tests one case where the false side is more numerous, but its extra wrong-side passages are generated for the same substitution, which a real attacker would vary, and include ConflictBank's temporal and semantic conflicts, which we count as misinformation. RAMDocs' document labels are imperfect, and its pair rule does not align entities (Section 4). We exclude 35 hold-out questions whose status depends on documents we could not label without human annotation, and check that conclusions survive treating those documents as misinformation. RAGuard's misleading side is real text, but its labels are behavioural (whether GPT-4 judged the claim correctly with the post). An LLM audit finds only 6% of sampled supporting–misleading pairs to contradict (Appendix D), so RAGuard's oracle is a label transfer, not a factual oracle. With posts in context, generators abstain on 32.1–60.9% of conflict claims, so accuracy on this true/false task falls below chance; on answered claims it is 53.8–68.4%, against 64.1–74.1% closed-book. Every generator is more accurate closed-book than with RAGuard's posts (gpt-4o-mini 64.9% against 36.7%), so the posts make a weak retrieval setting. HealthContradict's pages are real, but its labels are as released: an LLM judge finds the correct page supporting the gold answer for 68% and the misinformation page the wrong answer for 55%. WikiContradict has genuine conflicts but no misinformation side. CONFLICTS shows no significant effect for any condition; scored only on items with a reference answer, it can neither confirm nor refute our findings. Our comparisons detect effects of about 5.8 points with 80% power, so smaller effects, including small harms on clean questions, may be missed.

Metric. RAMDocs ships no scorer. Our primary metric reconstructs the published definition and scores our MADAM-RAG port close to the published number on the pilot split; as the published subset is unknown, this is not a reproduction. The metric is string matching: it misses paraphrases, and partial answers interact with "no wrong answer". We report the more lenient bidirectional variant throughout and mark conclusions depending on it. An outside-family LLM judge agrees with the matcher on 92.8% of 2400 paired outputs. The matcher's misses depend on the condition (more under answer keys and directional flags) but move no contrast by more than $+3.3$

points (Appendix D). Under that judge's re-score of all 23,204 responses, neither pooled equivalence holds within ± 3 (the flag leads by $+2.4$ and $+2.5$; H38), so those claims hold under the matcher only. No human labels check either scorer: the paired, blinded audit sheet we release is unlabelled.

Direction and the grid. Oracle direction is close to an answer key (H23), so oracle rows are ceilings, not findings about detectors. Direction precision is the share of a detector's flags on true contradictions naming the false side; pair precision, the share of its flagged pairs that are true contradictions. The gpt-4o-mini LLM detector's direction precision is 88.3% on RAMDocs and 68.8% on ConflictBank. Its per-question hit rate is lower because it also misses pairs. Its directional flag's gain over all ConflictBank questions with gpt-4o-mini is not significant (H15), and with stronger generators it hurts. The grid's gain rises with direction accuracy, but the grid cannot separate adjacent levels. Synthetic false flags are spread uniformly with a random suspect side; real detectors concentrate their errors. The grid predicts real detectors only roughly, and three detectors cannot show whether document- or pair-level summaries predict better.

Pre-registration. H1–H9 were registered before the confirmatory run and H10–H38 in nine addenda, after earlier results were known but before their own runs. The only timestamp is the repository's commit history, not a third-party registry. Holm controls error within each family, not across the programme: of 532 registered tests, 276 are significant under Holm and 304 under a Benjamini–Hochberg FDR over all tests (both in the hypothesis tables). Several analyses, labelled as such, were chosen after seeing data: the TOST margin, the single- and multi-answer split, the document-level placement axes, the post-hoc comparison of detector and generator reliability that H24 later tested, H24's mechanism score, and the refinement of permutation p-values at their floor.

Scope. We study flags passed to the generator as text. Learned interfaces such as soft prompts or credibility fine-tuning (Ye et al., 2026; Pan et al., 2024), and retrieval-side filtering, may trade off differently. We test one adversarial case, a detector that always names the wrong side, not an attacker who chooses the flagged pairs.

Ethics Statement

This work uses public benchmarks and commercial model APIs, with no human subjects. RAGuard's passages are public Reddit posts that can contain usernames; we use them as released, and our released RAGuard audit sheet replaces private users' handles with placeholders. The main risk is dual use: a false "likely misinformation" label, or removing a passage, can change a RAG system's answers. An inverted oracle that names the correct side as misinformation costs 7.0 points on RAMDocs and 8.6 on ConflictBank as a flag, and 21.8 on RAMDocs when the passage it names is removed. An adversary who can trigger a deployed detector could exploit this (Zou et al., 2025). We therefore report harm on clean questions alongside every gain, and argue that systems should measure a detector's document-level behaviour before acting on its flags.

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A Prompts

The prompts are verbatim from the code that produced the results. The examples use placeholder passages; in the runs every condition except removal shows the generator the same passages (checked against the run records).

Generation skeleton (default).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Minimal skeleton (no answer policy).

```
Answer the question using the passages below.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your answer; if there is more than
one, separate them with |>
```

Single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Closed book (single-answer version).

```
Answer the question from your own knowledge. The
question has exactly one correct answer.
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Instruction v1.

```
Some passages may contradict each other. Before
answering, check whether they do.
Ignore passages that appear to contain
misinformation, and if a disagreement reflects
genuinely different valid answers, report each one.
```

Instruction v2.

```
Some passages may contain false information. Before
answering, decide which passages are reliable by
checking them against each other.
Do not include an answer that is supported only by a
passage you judge to be unreliable.
```

Single call (inserted as the instruction).

```
Some passages may contain misinformation. First
decide which passages, if any, contain
misinformation, by checking them against each other
and against what you know, and write one line in
this format:
Suspect passages: <passage numbers, or none>
Then answer the question using only the other
passages.
```

Example: directional flag (passage 2 suspect).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION; conflicts with
passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; passage
2 likely contains misinformation

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Example: one-wrong flag, single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [CONFLICT with passage(s): 2; one side
contains misinformation]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [CONFLICT with passage(s): 0; one side
contains misinformation]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; exactly
one of the two contains misinformation (it is not
known which)

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Directional LLM detector.

```
You will see a question and {n} numbered passages
retrieved for it.
Some passages may contain misinformation: false
claims about the answer to the question.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

For each contradiction, say which passage of the
pair is more likely the misinformation, judging
by agreement with the other passages and by
plausibility: its number, "both" if both are likely
false, or "unclear" if you cannot tell.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation":
"contradiction|ambiguity|complementary|temporal",
"misinformation": <int>|"both"|"unclear"}}]}}
Use the passage numbers shown. "misinformation" is
needed only for contradictions. Return {{"pairs":
[]}} if there are none.
```

LLM detector without direction.

```
You will see a question and {n} numbered passages
retrieved for it.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
```

```
<int>, "relation": "contradiction|ambiguity|complementary|temporal"}}]}}
Use the passage numbers shown. Return {{"pairs": []}} if there are none.
```

Forced-choice skeleton (no "unknown").

```
Answer the question using the passages below. The question has exactly one correct answer.
You must give your best answer; "unknown" is not allowed.
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in exactly this format:
Final answer(s): <your one answer>
```

Pick instruction (with the neutral flag).

```
When flagged passages conflict, decide which side is better supported by the other passages and by what you know, and answer with that side only.
```

Example: neutral flag, single-answer skeleton.

```
Answer the question using the passages below. The question has exactly one correct answer.
If no passage answers the question, the answer is "unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [POTENTIAL CONFLICT with passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction

Question: Who founded X?

Give a brief explanation, then end with one line in exactly this format:
Final answer(s): <your one answer>
```

Example: neutral flag without pairs.

```
Answer the question using the passages below. The question has exactly one correct answer.
If no passage answers the question, the answer is "unknown".

Passage 0 [POTENTIAL CONFLICT]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [POTENTIAL CONFLICT]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Some of the marked passages conflict with each other.

Question: Who founded X?

Give a brief explanation, then end with one line in exactly this format:
Final answer(s): <your one answer>
```

Example: one-wrong flag without pairs.

```
Answer the question using the passages below. The question has exactly one correct answer.
If no passage answers the question, the answer is "unknown".

Passage 0 [IN CONFLICT; some marked passages contain misinformation]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [IN CONFLICT; some marked passages contain misinformation]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Some of the marked passages contain misinformation (it is not known which).

Question: Who founded X?

Give a brief explanation, then end with one line in exactly this format:
Final answer(s): <your one answer>
```

Example: directional flag without pairs.

```
Answer the question using the passages below. The question has exactly one correct answer.
If no passage answers the question, the answer is "unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Passages marked LIKELY MISINFORMATION probably contain false information.

Question: Who founded X?

Give a brief explanation, then end with one line in exactly this format:
Final answer(s): <your one answer>
```

Example: answer key (also the verdict key, with the detector's answers).

```
Answer the question using the passages below. The question has exactly one correct answer.
If no passage answers the question, the answer is "unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

Passage 2:
<passage 2>
```

```
NOTE: The answer "<wrong answer>" is false.

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: hint (the detector model's own answer).

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

NOTE: Another model that read these passages
answered: "<other model's answer>".

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: WikiContradict (default skeleton, neutral flag).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 1]:
<passage 0>

Passage 1 [POTENTIAL CONFLICT with passage(s): 0]:
<passage 1>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 1: possible contradiction

Question: <question>

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Paraphrase prompt (artefact probe, H29).

```
Paraphrase the passage below. Keep every fact, name,
number, date and place exactly as stated, including
any that seem wrong; do not add, remove or correct
information, and keep about the same length. Output
only the paraphrased passage.

Passage:
{passage}
```

LLM judge of the string matcher.

```
You are checking how a model's answer was scored.
Read the question, the correct answer(s), the wrong
(misinformation) answer(s), and the model's full
response. Judge only the model's final answer.

Question: {question}
Correct answer(s): {gold}
Wrong answer(s): {wrong}

Model response:
<<<
{response}
>>>

Reply with JSON only: {{"gives_all_correct":
true/false, "gives_wrong": true/false, "abstains":
true/false}}
- gives_all_correct: the final answer states every
correct answer (paraphrases and aliases count).
- gives_wrong: the final answer states any wrong
answer as an answer (mentioning it only to reject it
does not count).
- abstains: the final answer is "unknown" or
declines to answer.
```

B Additional Results

Published baselines (H14). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) lose same-name answers on RAMDocs (H14). On ConflictBank, the registered H14 compared our pipeline's multi-answer prompt with the baselines' own one-answer prompts. Under the matched one-answer prompt (our principal comparison), our pipeline beats them by +9.2 and +13.4 points (exploratory). Our TrustRAG port first omitted the repository's system message; sending it (§17.2) changes TrustRAG's accuracy by −1.0 [−3.4, +1.3] points on ConflictBank, −0.5 [−2.8, +1.8] under majority misinformation and −1.0 [−3.0, +0.9] on RAMDocs (−0.4 [−2.0, +1.4] for its multi-answer variant). Against the corrected TrustRAG, with policies matched (Holm over the four), the LLM directional flag changes accuracy by +14.4 [+9.3, +19.4] on ConflictBank (one-answer), +23.4 [+18.5, +28.7] on RAMDocs (multi-answer), −0.5 [−5.5, +4.6] under majority misinformation, and +1.8 [−2.8, +6.5] under the registered mismatched prompts. On ConflictBank, NLI first saw passage prefixes that usually omit the changed fact. Cutting each passage to the sentences closest to the question, up to the registered 1,500 characters, keeps both answers, but the pair still exceeds the cross-encoder's 512 tokens and NLI flags no pair (pair recall 0.0%). With 300 characters, about one sentence (a disclosed deviation, §17.6), NLI finds 21.0% of the gold pairs at 24.7% pair precision and flags 36.0% of clean questions. Its flag changes conflict accuracy by −0.2 [−1.9, +1.3] points (removal: +7.6 [+4.1, +11.2]). A single call that names suspect

passages before answering does not differ significantly from the pipeline (H18; Table 35).

Robustness to the metric and the labels. Table 29 repeats Table 6 with the bidirectional metric, and Table 30 repeats the pre-registered tests treating every suspect document as misinformation. Table 7 lists all pre-registered tests.

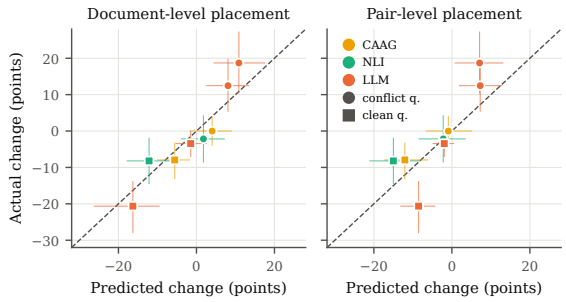


Figure 2: Effect of real detectors predicted from the synthetic grid (x, with the 95% bootstrap prediction interval) against the measured effect (y, with its 95% CI), for conflict (●) and clean (■) questions. Left: detectors located by document-level behaviour; right: by pair precision, recall and direction accuracy.

The heuristic detector. CAAG, the detector we set out to study, scores a pair only if its MiniLM (Reimers and Gurevych, 2019) cosine similarity reaches $\theta_s = 0.65$. Its score is the sum of a negation and contrast asymmetry term, a named-entity mismatch term (differing dates, numbers, money or percentages) and a temporal-marker term, multiplied by the similarity; a pair is flagged at $\theta_c = 0.35$. Its per-pair flag rate on the confirmatory split (6.22%) sets the random detector's rate. As a neutral flag, CAAG did not beat random flags of the same volume, and as a removal rule it only deleted correct passages (-7.9 points on clean questions, 0.0 on conflict ones). Its false flags mostly hit pairs that should both be reported: 134 of 221 on the hold-out are same-name ambiguity pairs, flagged by its entity-mismatch signal for differing dates or numbers. We read this as a warning about lexical contradiction cues generally (Gururangan et al., 2018), not a particular threshold. Retuning shows why F1 misleads: over a 121-setting sweep, CAAG's best pair F1 is 29.2, above NLI's, at $\theta_s = 0.40$ and $\theta_c = 0.10$, but it flags 43% of all pairs and 87% of clean questions, which our grid predicts is harmful under removal (Appendix G).

Original CAAG format. CAAG's flags in their original format, with typed lines and confidence scores ("factual conflict (confidence: 0.41)"), change accuracy by -0.9 points overall (-1.2 on conflict questions, -1.6 on clean ones), no different from the uniform format.

Pilot study (exploratory). Before any confirmatory run, the 200-question pilot split, scored with the bidirectional metric, showed the same pattern: oracle neutral flags changed conflict-question accuracy by -10.5 points, directional flags by $+11.8$, and removing the suspect by $+28.4$. The pilots also showed that the original REMOVE strategy, which deletes an arbitrary side of each pair, removes correct documents, and led us to add directional formats, synthetic direction accuracy and document-level placement. All pilot runs are released.

Phase-0 audit of CAAG on NarrativeQA. CAAG was designed for long-document QA. On 100 NarrativeQA (Kočiský et al., 2018) questions from 50 stories (BM25 top-10 chunks from a single coherent story), it flags at least one pair in 54% of questions, 166 pairs in all. A single story rarely contradicts itself, so most flags are likely false. The flagged pairs are released for annotation.

C Further Results

C.1 Where the directional flag fails (exploratory)

After seeing the minimal-skeleton run, we split conflict questions by gold-answer count. 105 of 139 have several (same-name ambiguity *and* misinformation), and a strict answer must list all. On the 34 single-answer conflict questions, the directional flag's estimate is positive in every setting, from $+11.8$ points (Qwen) to $+46.1$ (gpt-4o-mini), both with the default skeleton. On multi-answer questions it changes little except with gpt-4.1 (gpt-4o-mini $+2.9$, and -5.7 with the minimal skeleton; gpt-4.1 $+19.0$). One untested reading: a warning about one passage makes the smaller models more conservative about legitimate alternatives too. Removal, which needs no warning to be acted on, gains in both groups.

C.2 Real detectors

CAAG flags 34.1% of clean questions. Its neutral flags do not beat random flags at the same rate (H5; -0.4 vs. $+1.0$ points overall). The 90% CI of the difference, $[-3.1, +0.1]$, rules out a gain

Claim (section)	Evidence	Status
A correct neutral flag lowers RAMDocs accuracy (§5.1)	H2, H9	registered before any confirmatory data
Its mechanism depends on the answer policy: added misinformation, abstention, little effect (§5.1)	Table 2; H15, H19–H21, H25, H26	answer mix: re-analysis; tests: registered addenda
Pooled effect, policy split, moderators, utility, transitions (§5.1)	Tables 38, 39	re-analysis of data already seen; dependence: §17
The directional flag beats the neutral one (§5.2)	H3, H9–H12, H15, H16, H19, H25, H26	H3: before any confirmatory data; rest: addenda
Oracle direction is close to an answer key (§5.2)	H23 and its TOST	registered addendum; TOST registered in §13.4
Pooled, oracle direction is equivalent to an answer key on average, under the matcher (§5.2)	pooled H23; per-comparison TOST	unregistered re-analysis; heterogeneity: §17
Under the judge, neither pooled equivalence holds (§5.2, §5.3)	H38	registered (§19), LLM judge
Direction helps and a neutral flag does not on real web pages (§5.2)	H37	registered (§19); labels as released
Real-detector gains depend on the generator (§5.3)	H15, H17, H19, H25, H26	registered addenda
Right flags gain less than wrong flags cost; similar follow rates (§5.3)	H31	registered addendum (§16)
Break-even direction precision rises with generator accuracy (§5.3)	Table 25	unregistered re-analysis
Break-even is a calibration, not a rule for selective detectors (§5.3)	Table 42	re-analysis (§17)
Follow rates and coverage account for the crossed cells (§5.3)	Figure 1	unregistered; retrospective
Frozen, the account predicts unseen ConflictBank cells; it overstates harm on clean questions and HealthContradict (§5.3)	H32–H36, Table 43	registered (§19), predictions committed first
Registered reliability gap correlates with the gain (§5.3)	H24	registered addendum (§13), 5 of 30 cells existed
The gap is dominated by generator accuracy; precision and recall matter within generators (§5.3)	Table 37	re-analysis of data already seen
The detector's verdict as an answer key matches the flag; its model's answer as a hint does not (§5.3)	H30, H27	registered addenda (§15, §16); pooled TOST unregistered
A real detector's flag hurts when conflicts are rare (§5.3)	Table 26	unregistered re-analysis
Direction survives reordering and paraphrase (§5.3)	H29	registered addendum; equivalence not shown
Direction and removal harm genuine conflicts (§5.3)	H28	registered addendum
The matcher's errors do not overturn the directional conclusions (App. D)	Table 40	registered re-score (§17), LLM judge
Recommendations (§6)	–	not tested end to end

Table 3: Status of each claim. *Registered addendum*: committed before the runs it covers but after earlier results were known. *Re-analysis*: registered after the data were seen, or exploratory. The commit history is the only timestamp.

above +0.1 points but does not establish equivalence within ±3 points (TOST). The addendum registered the LLM detector's directional-versus-neutral comparison as exploratory. Under the bidirectional metric, the LLM detector's directional flag costs clean questions 6.1 points. With gpt-4.1 generating, its gain over all 300 questions (+5.4) is significant after Holm, but the 35 unlabelled questions contribute disproportionately, gaining 13.3 points. On labelled questions it is +4.4 [+0.6, +8.3]. On ConflictBank the NLI cross-encoder flags no pair: the benchmark's passages are long, LLM-written and agree on all but the one fact, and the truncated pair rarely contains it.

Stronger models abstain in the single call. Asked to name suspect passages before answering, the reasoning models and Claude Haiku distrust most of ConflictBank's generated passages and abstain, losing up to 51.0 points on clean questions (Table 35).

Reasoning models on RAMDocs. Without flags, gpt-6-luna and gpt-6.1-terra are strictly correct on few RAMDocs conflict questions because the default prompt asks for every valid answer. They give 2.8 and 3.0 answers on average and recover 79% and 77% of gold answers, but 72% and 76% of their answers include the misinformation.

Inverted oracle (names the correct side)	Conflict	Clean
RAMDocs, directional flag	−7.0 [−11.5, −2.9]	0.0
RAMDocs, removal	−21.8 [−28.8, −15.3]	0.0
ConflictBank (multi-answer), directional flag	−8.6	0.0

Table 4: Adversarial control, gpt-4o-mini: change from no flags, in points (exploratory; §9.5). On clean questions the inverted oracle flags nothing (prompt unchanged).

Detector parsing. The LLM detectors' JSON failed to parse on 1 of 2400 ConflictBank questions (a Claude Sonnet output), counted as flagging nothing.

C.3 A detector from another family

Replacing the gpt-4o-mini detector with Qwen2.5-7B (gpt-4o-mini still generates) gives a much less selective detector, flagging 96.8% of clean RAMDocs questions, but with direction accuracy 89.0%. Its directional flag changes accuracy by +10.6 points on conflict questions and 0.0 on clean ones; its neutral flag, by −2.6. Removal again trades a conflict-question gain (+14.1) for a clean-question loss (−14.6). The conclusions do not depend on the detector's family.

C.4 CONFLICTS

On the 237 automatically scorable CONFLICTS questions (exploratory, 3 seeds; most have no conflict), no condition moves accuracy significantly (Table 32).

D Semantic Re-score, Label Audits and Selection

An outside-family judge leaves the directional conclusions intact (§17.3). To check whether the matcher's errors differ by condition, an outside-family LLM re-scored both arms of 10 contrasts on the same 120 conflict questions each (gemini-3.5-flash-lite; Table 40). §17.3 named Gemini 2.5 Flash, closed to new users. Given the gold and wrong answers, the judge tests the scorer, not the benchmark labels. Judge and matcher agree on 92.8% of 2400 outputs. Almost all disagreements are answers the judge accepts and the matcher rejects (aliases, paraphrases, partial forms). Their share depends on condition: 10.3% of answer-key outputs and 11.7% of oracle-directional ones, against 5.2% without a flag. The failure of most concern, a final answer such as "Y, not X" counted as asserting the wrong answer X, occurs in 0.2% of outputs. Since both arms gain, the judge moves every contrast by −1.7 to +3.3 points. One small contrast changes sign (RAMDocs oracle flag against answer key, −0.8 [−5.8, +4.2] to +0.8 [−8.3, +10.0]). Two whose matcher intervals include zero exclude it under the judge (unadjusted): Claude Haiku's neutral flag (+6.7 [+0.8, +13.3]) and gpt-4o-mini's LLM flag against its verdict key (+3.3 [−2.5, +9.2] to +6.7 [+0.8, +12.5], outside ±3). With 120 questions per contrast the re-score's intervals are too wide to test the ±3-point equivalence claims; re-scored in full, neither holds (Appendix E). A blinded, paired human-audit sheet of 150 of these pairs and 50 RAGuard post pairs is released unlabelled (`runs/labelling/audit8_*`).

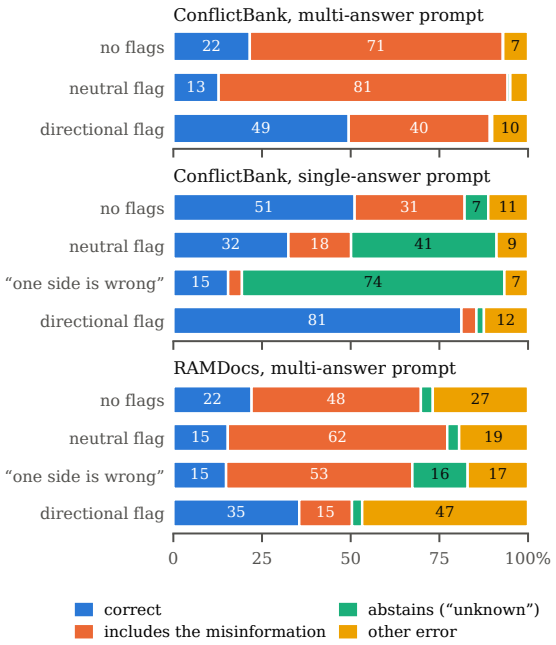


Figure 3: What a flag does to answers on conflict questions (gpt-4o-mini, oracle pairs; share of answers, %). The answer policy decides how a neutral flag fails: told to give every valid answer, a model adds the misinformation; told there is one answer, it abstains. Only the directional flag raises the share of correct answers. Categories are exclusive, correct (under the primary metric) taking precedence, so the misinformation share can be slightly lower than in Table 2, which counts every answer containing a wrong answer under the bidirectional matcher.

			Accuracy		Oracle pairs				LLM detector		Single	LLM dir.
Generator	Prompt	Seeds	Closed	Plain	Flag	One wrong	Dir. flag	Remove	Flag	Dir. flag	call	clean q.
gpt-4o-mini	single	3	10.0	51.1	**−18.7**	**−35.7**	**+30.1**	**+35.0**	**−18.2**	+8.4	+7.6	−4.7
gpt-4o-mini	multi	3	–	21.6	**−8.8**	–	**+27.9**	**+66.4**	**−8.4**	**+20.0**	–	−1.0
gpt-4o-mini	minimal	3	–	35.0	**−18.1**	–	**+38.4**	**+51.2**	–	–	–	–
gpt-4.1	single	1	20.0	64.7	**−8.0**	**−35.3**	**+26.7**	**+26.0**	–	−2.3	+7.3	−3.0
gpt-4.1	multi	1	–	13.0	**−6.3**	–	**+23.0**	**+78.3**	**−6.7**	**+17.0**	–	0.0
Qwen2.5-7B	single	1	4.3	50.3	−3.3	−5.0	**+25.3**	**+35.7**	−4.0	+7.3	–	−1.0
Qwen2.5-7B	multi	1	–	23.7	**−8.3**	–	**+17.7**	**+63.7**	−8.3	+6.7	–	−1.0
gpt-6-luna	single	3	16.9	53.6	**−9.4**	**−29.1**	**+36.4**	**+37.0**	**−9.3**	+8.6	−6.3	−1.3
gpt-6.1-terra	single	1	20.0	73.0	−2.3	−5.3	**+17.0**	**+18.3**	−2.7	−11.7	**−21.0**	0.0
Claude Haiku 5.5	single	3	11.8	66.8	+0.7	+6.6	**+23.3**	**+22.6**	+0.7	−4.1	+1.7	−5.0
Claude Sonnet 5.5	single	1	–	75.7	+3.0	+4.0	**+16.3**	–	–	−2.7	–	−1.0
gpt-4o-mini	forced	3	–	55.4	−0.2	**+7.6**	**+24.2**	–	–	+4.0	–	−2.7
gpt-6-luna	forced	3	–	64.6	+1.6	**+10.3**	**+26.4**	–	–	−2.4	–	−1.3
Claude Haiku 5.5	forced	3	–	69.3	**+3.9**	**+5.1**	**+21.1**	–	–	−4.7	–	−1.3
gpt-6.1-terra	forced	1	–	77.7	−0.3	**+6.7**	**+14.7**	–	–	**−14.3**	–	0.0
Majority misinformation: three wrong-side passages, one correct												
gpt-4o-mini	single	3	–	21.6	+1.1	−5.0	**+32.7**	**+65.1**	−2.2	**+5.7**	+6.8	−4.7
gpt-6-luna	single	3	–	29.8	**+6.6**	+0.3	**+60.4**	–	–	**+7.1**	–	−1.3
gpt-6.1-terra	single	1	–	26.0	−3.7	**+22.0**	**+66.3**	–	–	**+9.0**	–	0.0
Claude Haiku 5.5	single	3	–	31.0	+4.3	–	**+57.3**	–	–	**+6.8**	–	−5.0
RAGuard: real Reddit posts labelled supporting or misleading (294 conflict claims)												
gpt-4o-mini	single	3	64.9	36.7	+3.2	−5.7	**+9.9**	+5.9	–	+0.6	–	+1.7
gpt-6-luna	single	3	64.1	23.2	0.0	**−7.0**	**+8.8**	−1.0	–	+3.1	–	−1.3
Claude Haiku 5.5	single	3	65.1	24.5	+0.3	**+5.9**	**+12.7**	+2.0	–	**+5.0**	–	+1.3
gpt-6.1-terra	single	1	74.1	24.8	+2.4	+0.7	**+6.8**	−1.0	–	+2.0	–	+1.0
Claude Sonnet 5.5	single	1	–	27.2	+4.4	–	**+20.4**	–	–	**+7.1**	–	+10.0

Table 5: ConflictBank conflict questions (300; one correct answer each) and RAGuard conflict claims (294; answer true or false): accuracy (%, one-way strict) closed-book and with the passages but no flags, then the paired difference in points made by each format. *forced*: the one-answer prompt without the option to answer "unknown". *Prompt*: the answer policy: *single* says there is one answer, *multi* asks for every valid answer (our default skeleton), *minimal* states no policy. *One wrong*: the neutral flag plus "exactly one of the two contains misinformation". *Single call*: one call that names the suspect passages, then answers. Last column: the LLM directional flag's change on the 100 clean questions. The LLM detector is gpt-4o-mini for every generator. **Bold**: significant under the paper's single scheme.

The neutral-flag cost holds without cross-entity pairs (§18). RAMDocs' pair rule labels a pair a contradiction whenever one side is misinformation and the answers differ, without aligning entities. Assuming each misinformation passage targets one entity, at least 261 of the 709 oracle contradiction pairs on the hold-out join it to a correct passage about another entity, in the 98 questions whose correct passages give several answers. On the 41 single-entity questions no pair can cross entities. Averaged over the 10 RAMDocs settings, the oracle neutral flag changes accuracy by −4.4 [−7.3, −1.5] points there and −4.7 [−6.7, −2.7] on the rest; the oracle directional flag by +29.2 [+21.0, +37.8] and +12.9 [+9.0, +17.1] (Table 41). Per question, the flag and its key disagree on 9.9% (H23; flag only 4.9%, key only 5.0%) and 9.4% (H30; 5.3% and 4.1%) of question–seed pairs. On the cells with two no-flag seeds, the seeds disagree on 6.7% and 9.5%, and flag and key on 11.7% and 8.3%.

The WikiContradict harm holds under its original rubric. The original rubric counts an answer correct when it gives both answers, acknowledges the conflict and does not prefer one without support (Hou et al., 2024, §3.4). Under it, the LLM directional flag changes the share of correct answers by −6.7 [−11.9, −1.6] points for gpt-4o-mini (16.2% to 9.5%), −31.6 [−37.9, −25.3] for gpt-6-luna and −28.1 [−34.0, −22.1] for Claude Haiku (H28; seed 0, all 253 questions). For gpt-4o-mini the flag raises the share that merely mentions both answers (+5.1 [0.0, +10.3]) while lowering the share acknowledging the conflict without an unsupported preference.

RAGuard pairs rarely contradict. RAGuard labels a post supporting or misleading by whether GPT-4 judged the claim correctly with it in context, so its "contradiction pairs" transfer behavioural labels. Of 150 sampled supporting × misleading pairs, the judge finds the two

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	29.9	31.3	22.1	41.5	911
Prompt only					
Instruction (v1)	+1.6	+0.8	−1.2	+2.9	948
Instruction (v2)	+3.1	+2.3	+5.3	−1.1	951
Oracle detector					
Oracle, flag	−3.1	−3.5	**−6.7**	0.0	980
Oracle, directional flag	+6.2	+7.0	**+13.4**	0.0	998
Oracle, remove suspect	**+12.8**	**+14.5**	**+27.6**	0.0	825
Real detectors (exploratory)					
Random, flag	+1.0	+1.0	+0.7	+1.3	949
CAAG, flag	−0.4	−0.8	−0.7	−0.8	942
CAAG, remove	−3.3	−3.8	0.0	−7.9	809
NLI, flag	+0.2	+0.3	−0.5	+1.1	1021
NLI, remove	−5.7	−5.0	−2.2	−8.2	592
LLM, flag	−1.8	−1.3	−4.1	+1.9	970
LLM, directional flag	+5.0	+4.9	+12.5	−3.4	983
LLM, remove suspect	+0.3	0.0	+18.7	**−20.6**	718
Reference					
MADAM-RAG	+5.7	+5.7	+3.6	+7.9	8196

Table 6: RAMDocs hold-out, gpt-4o-mini, 3 seeds, one-way strict accuracy (primary). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

posts cannot both be right about the claim in 6%, the misleading post false about it in 21% (unclear in 57%) and the supporting post consistent with the verdict in 71%; all three hold for 5%. Most pairs discuss the claim from different angles without asserting opposite facts. RAGuard's oracle-flag rows therefore test what a generator does with these labels, not verified contradictions; we draw no factual-oracle conclusion from them. This audit uses one LLM judge; the released human sheet covers 50 of these pairs.

Selection in the crossed cells. Table 42 splits each crossed cell by the flag's per-question effect. The marginal break-even calculation (precision times R plus its complement times W, at full coverage) has mean error 3.5 points against the observed gains (1.9 when scaled by recall); the follow-rate account, using each cell's coverage and no-flag accuracy, has 1.5 [1.4, 3.1].

E Prospective Test, Full Re-score and HealthContradict (§19)

Frozen, the account predicts unseen cells (H32–H36). ConflictBank-fresh has 400 questions built like our ConflictBank sample, with no shared subject. After detections and no-flag answers, we committed the follow-rate account's predictions, from H31's follow rates fixed in §19.1, (`splits/predictions19.json`) before running the flagged condition; the analysis reproduces them exactly (maximum drift 0.00 points). All three H32 criteria hold: the slope's CI [0.77, 1.00] excludes 0 and includes 1 (intercept +1.0); the mean error is $1.4 \leq 3$, and the account beats the generator-mean (4.1) and unit-follow-rate (5.3) baselines (Holm $p < 0.001$). A cell-level marginal estimate scores 1.8; predicting zero, 8.7. The sign is right in 20 of 20 cells and in all 18 with $|\text{pred}| \geq 3$. Of the 2 cells predicted to lose at least 3 points, 1 does so significantly, so H33 fails. The within-generator correlation is 0.97 (H34, $p < 0.001$). On clean questions (H35) the account predicts −4.9 points on average against −1.4 observed (mean error 3.6): a flag there changes fewer correct answers than H31's wrong-flag follow rate implies. A fresh randomised-direction run gives gpt-4o-mini follow rates of 50% [40, 61] and 50% [38, 60], inside the frozen intervals (H36). For every generator, the Claude Haiku detector has the highest predicted and observed effect (effects over all 400 questions, against the gpt-4o-mini detector: gpt-4o-mini generator +12.0 vs +7.7 points; gpt-6.1-terra +5.3 vs −3.5).

Under the judge, neither pooled equivalence holds (H38). The §17.3 judge scored every response in the pooled H30 and H23 tests (23,204 rows; agreement with the matcher 94.3%). On the same cells and pooling, the LLM flag minus the verdict key is +1.2 [+0.1, +2.2] (90% CI) under the matcher and +2.4 [+1.3, +3.5] under the judge; the oracle flag minus the answer key is −0.1 [−1.3, +1.0] and +2.5 [+1.1, +3.9]. Neither is equivalent within ±3 under the judge; the smallest passing margins are 3.5 and 3.9. Missing more correct answers under the keys, the matcher understates the flag's advantage by 1.2 and 2.6 points (matcher minus judge −1.2 [−1.9, −0.6] and −2.6 [−3.9, −1.4]). Under either scorer, both point estimates stay below 3 points.

Direction helps on real web pages (H37). The oracle directional flag beats no flag (H37a) and the neutral flag (H37c) for all three generators. The neutral flag changes accuracy by at most 2.4 points (H37b; within ±3 for gpt-6-luna and Claude Haiku). For every generator, closed-book accu-

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H1: detection quality					
H1	F1 llm_dir > nli	+2.6 [−2.4, +7.5]	0.311 (0.414)	not supported	–
H1	F1 nli > caag	+12.2 [+7.5, +16.9]	<0.001 (<0.001)	supported	–
H1	direction accuracy llm_dir > 50% (cluster CI[r])	88.3% [83.3, 92.9]	<0.001 (<0.001)	supported	–
H2: a correct neutral flag					
H2	oracle+flag < plain (conflict)	−6.7 [−10.8, −3.1]	<0.001 (0.002)	supported	yes
H3: what direction adds					
H3	oracle+flag_directional > plain (conflict)	+13.4 [+6.7, +20.4]	<0.001 (<0.001)	supported	yes
H3	oracle+flag_directional > oracle+flag (conflict)	+20.1 [+13.9, +26.9]	<0.001 (<0.001)	supported	yes
H3	oracle+remove_suspect > plain (conflict)	+27.6 [+20.6, +35.0]	<0.001 (<0.001)	supported	yes
H4: false flags					
H4a	false-flag cells below plain on clean q., directional flag[r]	0 of 9 cells	Holm	not supported	yes
H4a	false-flag cells below plain on clean q., removal[r]	6 of 9 cells	Holm	supported	yes
H4b	harm grows with clean-q. exposure, Spearman over 18 cells[r]	$\rho = -0.39$ [−0.74, −0.12]	–	supported	yes
H4c	removal harms clean q. more than directional flag[r]	5 of 9 matched cells	Holm	count only	–
H5: the heuristic detector vs random					
H5	caag+flag vs random+flag (all)	−1.4 [−3.4, +0.3]	0.178 (0.257)	n.s.	yes
H5	equivalence within ±3 points (TOST)[r]	90% CI [−3.1, +0.1]	–	not equivalent	yes
H6: does the grid predict real detectors					
H6	actual effect inside the 95% interval of the document-level prediction[r]	7 of 8 (pair level: 6 of 8)	–	descriptive	–
H6	document-level error < pair-level error (paired, exact sign-flip)[r]	−1.2 [−3.6, +0.4]	0.414	not supported	yes
H7: prompt-only instructions					
H7	instruct vs plain (conflict)	−1.2 [−4.3, +1.7]	1.000 (0.636)	n.s.	yes
H7	instruct vs plain (clean)	+2.9 [0.0, +6.1]	0.280 (0.145)	n.s.	yes
H7	instruct_v2 vs plain (conflict)	+5.3 [+1.4, +9.6]	0.059 (0.027)	n.s.	yes
H7	instruct_v2 vs plain (clean)	−1.1 [−5.0, +2.6]	1.000 (0.787)	n.s.	yes
H9: gpt-4.1					
H9	gpt-4.1: oracle+flag < plain (conflict)	−8.6 [−13.2, −4.3]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+flag_directional > plain (conflict)	+18.2 [+12.0, +24.7]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+26.9 [+20.1, +33.6]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+remove_suspect > plain (conflict)	+30.2 [+22.5, +37.9]	<0.001 (<0.001)	supported	yes
H8: genuine conflicts (WikiContradict)					
H8	WikiContradict: instruct vs plain	+8.7 [+5.4, +12.3]	<0.001 (<0.001)	significant	–
H8	WikiContradict: instruct_v2 vs plain	−6.7 [−10.0, −3.7]	<0.001 (<0.001)	significant	–
H8	WikiContradict: oracle+flag vs plain	+6.5 [+3.0, +10.0]	<0.001 (<0.001)	supported	–
H8	WikiContradict: caag+flag vs plain	+0.7 [0.0, +1.6]	0.250 (0.345)	n.s.	–
H8	WikiContradict: nli+flag vs plain	+3.2 [+0.3, +6.1]	0.077 (0.066)	n.s.	–
H8	WikiContradict: llm_dir+flag vs plain	+6.2 [+2.8, +9.6]	<0.001 (<0.001)	significant	–

Table 7: Pre-registered tests registered before the confirmatory run (RAMDocs hold-out unless stated; one-way strict accuracy; differences in points). Rows marked [r] are re-analyses registered later, before they were computed. Directional predictions are *supported* or *not supported*; tests with no predicted sign are *significant* or *n.s.* *Robust*: the same outcome under the bidirectional metric (robustness to the RAMDocs label rule is in Table 30, where H4b's correlation weakens). p-values come from 10,000 sign-flip permutations, extended to 200,000 when none reaches the observed statistic, Holm-adjusted within the family; this is the significance used everywhere in the paper. In brackets: the Benjamini–Hochberg q over all registered tests of the programme (primary metric).

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H10: without the answer policy (minimal skeleton)					
H10	gpt-4o-mini: oracle+flag vs plain (conflict)	-5.3 [-9.4, -1.4]	0.042 (0.020)	significant	yes
H10	gpt-4o-mini: oracle+flag_directional > plain (conflict)	$+5.0$ [-1.7, $+11.8$]	0.314 (0.231)	not supported	yes
H10	gpt-4o-mini: oracle+flag_directional > oracle+flag (conflict)	$+10.3$ [$+4.8$, $+16.1$]	0.005 (0.001)	supported	yes
H10	gpt-4o-mini: oracle+remove_suspect > plain (conflict)	$+18.9$ [$+12.5$, $+25.9$]	<0.001 (<0.001)	supported	yes
H10	gpt-4.1: oracle+flag vs plain (conflict)	-11.0 [-17.5, -4.6]	0.010 (0.003)	significant	yes
H10	gpt-4.1: oracle+flag_directional > plain (conflict)	$+12.7$ [$+5.8$, $+19.7$]	0.005 (0.001)	supported	yes
H10	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	$+23.7$ [$+17.0$, $+30.7$]	<0.001 (<0.001)	supported	yes
H10	gpt-4.1: oracle+remove_suspect > plain (conflict)	$+14.9$ [$+8.6$, $+21.6$]	<0.001 (<0.001)	supported	yes
H10	Qwen2.5-7B: oracle+flag vs plain (conflict)	-5.8 [-10.8, -1.4]	0.117 (0.067)	n.s.	yes
H10	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+2.9$ [-3.6, $+9.4$]	0.509 (0.613)	not supported	yes
H10	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+8.6$ [$+2.9$, $+14.4$]	0.041 (0.016)	supported	yes
H10	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+10.8$ [$+4.3$, $+17.3$]	0.017 (0.006)	supported	yes
H11: an open-weights generator					
H11	Qwen2.5-7B: oracle+flag < plain (conflict)	-5.8 [-10.8, -0.7]	0.114 (0.092)	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+3.6$ [-0.7, $+8.6$]	0.224 (0.319)	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+9.4$ [$+5.0$, $+14.4$]	0.001 (0.001)	supported	yes
H11	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+18.0$ [$+10.8$, $+25.2$]	<0.001 (<0.001)	supported	yes

Table 8: Pre-registered tests added in the first addendum (§9; H10–H11), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H12: ConflictBank					
H12	gpt-4o-mini: neutral flag vs plain (conflict)	−8.8 [−12.8, −4.9]	<0.001 (<0.001)	significant	yes
H12	gpt-4o-mini: dir. flag > plain (conflict)	+27.9 [+22.8, +32.9]	<0.001 (<0.001)	supported	yes
H12	gpt-4o-mini: dir. flag > neutral flag (conflict)	+36.7 [+31.6, +41.9]	<0.001 (<0.001)	supported	yes
H12	gpt-4o-mini: removal > plain (conflict)	+66.4 [+61.1, +71.7]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: neutral flag vs plain (conflict)	−6.3 [−9.7, −3.3]	<0.001 (<0.001)	significant	yes
H12	gpt-4.1: dir. flag > plain (conflict)	+23.0 [+18.0, +28.3]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: dir. flag > neutral flag (conflict)	+29.3 [+24.0, +34.7]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: removal > plain (conflict)	+78.3 [+73.3, +83.0]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: neutral flag vs plain (conflict)	−8.3 [−12.3, −4.3]	<0.001 (<0.001)	significant	yes
H12	Qwen2.5-7B: dir. flag > plain (conflict)	+17.7 [+12.0, +23.3]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: dir. flag > neutral flag (conflict)	+26.0 [+20.7, +31.3]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: removal > plain (conflict)	+63.7 [+58.3, +69.0]	<0.001 (<0.001)	supported	yes
H13: neutral flag + resolution instruction					
H13	RAMDocs, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	−2.4 [−6.2, +1.2]	0.743 (0.352)	n.s.	yes
H13	RAMDocs, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+15.6 [+9.6, +21.8]	<0.001 (<0.001)	supported	yes
H13	RAMDocs, gpt-4.1: neutral flag + instruction vs plain (conflict)	+5.8 [−2.9, +14.4]	0.743 (0.344)	n.s.	yes
H13	RAMDocs, gpt-4.1: dir. flag + instruction > neutral flag + instruction (conflict)	+25.2 [+18.0, +32.4]	<0.001 (<0.001)	supported	yes
H13	ConflictBank, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	+1.6 [−2.8, +6.0]	0.743 (0.622)	n.s.	yes
H13	ConflictBank, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+62.2 [+57.1, +67.2]	<0.001 (<0.001)	supported	yes
H14: published baselines					
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs astute multi (labelled)	+10.8 [+5.9, +16.0]	<0.001 (<0.001)	significant	yes
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs trustrag multi (labelled)	+23.0 [+18.1, +28.3]	<0.001 (<0.001)	significant	yes
H14	RAMDocs, gpt-4.1: LLM dir. flag vs astute multi (labelled)	+6.4 [+0.4, +12.5]	0.151 (0.082)	n.s.	no
H14	RAMDocs, gpt-4.1: LLM dir. flag vs trustrag multi (labelled)	+19.6 [+13.6, +25.7]	<0.001 (<0.001)	significant	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs astute (all questions)	−3.4 [−8.1, +1.3]	0.324 (0.237)	n.s.	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs trustrag (all questions)	+0.8 [−3.9, +5.6]	0.761 (0.844)	n.s.	yes

Table 9: Pre-registered tests added in the second (§10; H12–H14), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H15: ConflictBank, single-answer prompt					
H15	ConflictBank, gpt-4o-mini: neutral flag vs plain (conflict)	−18.7 [−23.4, −14.0]	<0.001 (<0.001)	significant	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > neutral flag (conflict)	+48.8 [+43.6, +54.1]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > plain (conflict)	+30.1 [+25.1, +35.2]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4o-mini: LLM dir. flag > plain (all q.)	+5.2 [+1.0, +9.3]	0.065 (0.030)	not supported	no
H15	ConflictBank, gpt-4.1: neutral flag vs plain (conflict)	−8.0 [−12.7, −3.7]	0.006 (0.003)	significant	yes
H15	ConflictBank, gpt-4.1: dir. flag > neutral flag (conflict)	+34.7 [+29.0, +40.3]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4.1: dir. flag > plain (conflict)	+26.7 [+21.7, +31.7]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4.1: LLM dir. flag > plain (all q.)	−2.5 [−6.5, +1.5]	0.522 (0.357)	not supported	yes
H15	ConflictBank, Qwen2.5-7B: neutral flag vs plain (conflict)	−3.3 [−9.0, +2.0]	0.522 (0.392)	n.s.	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > neutral flag (conflict)	+28.7 [+23.3, +34.0]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > plain (conflict)	+25.3 [+19.7, +31.0]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, Qwen2.5-7B: LLM dir. flag > plain (all q.)	+5.2 [+0.5, +10.0]	0.130 (0.073)	not supported	yes
H16: information vs direction (one-wrong flag)					
H16	ConflictBank, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+65.8 [+60.6, +70.9]	<0.001 (<0.001)	supported	yes
H16	ConflictBank, gpt-4o-mini: one-wrong flag vs plain (conflict)	−35.7 [−41.1, −30.3]	<0.001 (<0.001)	significant	yes
H16	ConflictBank, gpt-4.1: dir. flag > one-wrong flag (conflict)	+62.0 [+56.0, +67.7]	<0.001 (<0.001)	supported	yes
H16	ConflictBank, gpt-4.1: one-wrong flag vs plain (conflict)	−35.3 [−41.3, −29.3]	<0.001 (<0.001)	significant	yes
H16	RAMDocs, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+20.6 [+14.4, +27.3]	<0.001 (<0.001)	supported	yes
H16	RAMDocs, gpt-4o-mini: one-wrong flag vs plain (conflict)	−7.2 [−12.2, −2.6]	0.006 (0.011)	significant	yes
H17: majority misinformation					
H17	CB majority, gpt-4o-mini: neutral flag vs plain (conflict)	+1.1 [−2.7, +4.9]	0.603 (0.702)	n.s.	yes
H17	CB majority, gpt-4o-mini: dir. flag > plain (conflict)	+32.7 [+27.3, +37.9]	<0.001 (<0.001)	supported	yes
H17	CB majority, gpt-4o-mini: LLM dir. flag vs plain (conflict)	+5.7 [+1.4, +9.9]	0.017 (0.016)	significant	yes
H18: a single-call baseline					
H18	RAMDocs, gpt-4o-mini: LLM dir. flag vs self-identify (labelled)	−1.3 [−4.8, +2.1]	1.000 (0.626)	n.s.	yes
H18	ConflictBank, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−0.2 [−4.2, +3.7]	1.000 (0.996)	n.s.	yes
H18	CB majority, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−1.8 [−4.6, +1.1]	0.730 (0.339)	n.s.	yes
H19: current reasoning models					
H19	ConflictBank, gpt-6-luna: neutral flag vs plain (conflict)	−9.4 [−12.9, −6.0]	<0.001 (<0.001)	significant	yes
H19	ConflictBank, gpt-6-luna: dir. flag > neutral flag (conflict)	+45.9 [+40.8, +50.9]	<0.001 (<0.001)	supported	yes

Table 10: Pre-registered tests added in the third and fourth (§11–§12; H15–H19) (part 1 of 2), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H19: current reasoning models					
H19	ConflictBank, gpt-6-luna: dir. flag > one-wrong flag (conflict)	+65.6 [+60.7, +70.1]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6-luna: LLM dir. flag > plain (all q.)	+6.1 [+2.1, +10.0]	0.019 (0.008)	supported	yes
H19	ConflictBank, gpt-6.1-terra: neutral flag vs plain (conflict)	−2.3 [−6.7, +2.0]	0.717 (0.469)	n.s.	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+19.3 [+14.7, +24.0]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > one-wrong flag (conflict)	+22.3 [+17.7, +27.3]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6.1-terra: LLM dir. flag > plain (all q.)	−8.8 [−13.2, −4.2]	0.002 (<0.001)	not supported	yes
H19	RAMDocs, gpt-6-luna: neutral flag vs plain (conflict)	−3.1 [−5.3, −1.4]	0.007 (0.003)	significant	yes
H19	RAMDocs, gpt-6-luna: dir. flag > neutral flag (conflict)	+42.2 [+34.5, +49.9]	<0.001 (<0.001)	supported	yes
H19	RAMDocs, gpt-6.1-terra: neutral flag vs plain (conflict)	−0.7 [−2.9, +1.4]	1.000 (1.000)	n.s.	yes
H19	RAMDocs, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+40.3 [+32.4, +48.2]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6-luna: dir. flag > plain (conflict)	+60.4 [+55.3, +65.4]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6-luna: LLM dir. flag vs plain (conflict)	+7.1 [+2.4, +11.8]	0.019 (0.008)	significant	yes
H19	CB majority, gpt-6.1-terra: dir. flag > plain (conflict)	+66.3 [+61.0, +71.7]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	+9.0 [+3.3, +14.7]	0.019 (0.008)	significant	yes

Table 11: Pre-registered tests added in the third and fourth (§11–§12; H15–H19) (part 2 of 2), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H20: forced choice (no "unknown")					
H20	ConflictBank forced, gpt-4o-mini: neutral flag vs plain (conflict)	-0.2 [-4.1, $+3.7$]	1.000 (1.000)	n.s.	yes
H20	ConflictBank forced, gpt-4o-mini: one-wrong flag vs plain (conflict)	$+7.6$ [$+2.7$, $+12.3$]	0.024 (0.007)	significant	yes
H20	ConflictBank forced, gpt-4o-mini: dir. flag > plain (conflict)	$+24.2$ [$+19.7$, $+29.0$]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, gpt-6-luna: neutral flag vs plain (conflict)	$+1.6$ [-1.6, $+4.8$]	1.000 (0.478)	n.s.	yes
H20	ConflictBank forced, gpt-6-luna: one-wrong flag vs plain (conflict)	$+10.3$ [$+5.6$, $+15.1$]	<0.001 (<0.001)	significant	yes
H20	ConflictBank forced, gpt-6-luna: dir. flag > plain (conflict)	$+26.4$ [$+21.9$, $+31.0$]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain (conflict)	-0.3 [-4.0, $+3.3$]	1.000 (1.000)	n.s.	yes
H20	ConflictBank forced, gpt-6.1-terra: one-wrong flag vs plain (conflict)	$+6.7$ [$+2.3$, $+11.3$]	0.034 (0.013)	significant	yes
H20	ConflictBank forced, gpt-6.1-terra: dir. flag > plain (conflict)	$+14.7$ [$+10.3$, $+19.0$]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain (conflict)	$+3.9$ [$+1.2$, $+6.7$]	0.034 (0.011)	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: one-wrong flag vs plain (conflict)	$+5.1$ [$+1.4$, $+8.8$]	0.034 (0.016)	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: dir. flag > plain (conflict)	$+21.1$ [$+16.9$, $+25.4$]	<0.001 (<0.001)	supported	yes
H22: pair topology					
H22	RAMDocs, gpt-4o-mini: one-wrong without pairs vs with pairs (conflict)	$+6.5$ [$+2.4$, $+10.8$]	0.015 (0.005)	significant	yes
H22	RAMDocs, gpt-6-luna: one-wrong without pairs vs with pairs (conflict)	$+10.1$ [$+5.8$, $+14.9$]	<0.001 (<0.001)	significant	yes
H22	RAMDocs, gpt-6.1-terra: one-wrong without pairs vs with pairs (conflict)	-0.7 [-6.5, $+5.8$]	1.000 (1.000)	n.s.	yes
H22	CB majority, gpt-4o-mini: neutral without pairs vs with pairs (conflict)	-3.1 [-6.4, $+0.2$]	0.390 (0.124)	n.s.	yes
H22	CB majority, gpt-4o-mini: LLM dir. flag without pairs vs with pairs (conflict)	$+0.3$ [-2.1, $+2.8$]	1.000 (0.935)	n.s.	yes
H22	CB majority, gpt-6-luna: neutral without pairs vs with pairs (conflict)	-6.1 [-9.1, -3.1]	0.002 (<0.001)	significant	yes
H22	CB majority, gpt-6-luna: LLM dir. flag without pairs vs with pairs (conflict)	$+0.8$ [-1.7, $+3.1$]	1.000 (0.683)	n.s.	yes
H22	CB majority, gpt-6.1-terra: neutral without pairs vs with pairs (conflict)	$+10.0$ [$+5.7$, $+14.3$]	0.002 (<0.001)	significant	yes
H22	CB majority, gpt-6.1-terra: LLM dir. flag without pairs vs with pairs (conflict)	$+1.7$ [-1.3, $+4.7$]	1.000 (0.474)	n.s.	yes
H23: answer key					
H23	RAMDocs, gpt-4o-mini: dir. flag vs answer key (conflict)	-1.7 [-6.2, $+3.1$]	1.000 (0.656)	n.s.	yes
H23	RAMDocs, gpt-6-luna: dir. flag vs answer key (conflict)	-4.1 [-9.1, $+0.7$]	0.808 (0.202)	n.s.	yes
H23	RAMDocs, gpt-6.1-terra: dir. flag vs answer key (conflict)	-5.0 [-10.8, $+0.7$]	0.827 (0.241)	n.s.	yes
H23	ConflictBank, gpt-4o-mini: dir. flag vs answer key (conflict)	$+5.3$ [$+1.6$, $+9.1$]	0.059 (0.013)	n.s.	no
H23	ConflictBank, gpt-6-luna: dir. flag vs answer key (conflict)	$+0.1$ [-1.6, $+1.8$]	1.000 (1.000)	n.s.	yes

Table 12: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 1 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H23: answer key					
H23	ConflictBank, gpt-6.1-terra: dir. flag vs answer key (conflict)	-1.7 [-3.3, -0.3]	0.515 (0.104)	n.s.	yes
H23	CB majority, gpt-4o-mini: dir. flag vs answer key (conflict)	$+3.2$ [-1.8, $+8.2$]	0.905 (0.319)	n.s.	yes
H23	CB majority, gpt-6-luna: dir. flag vs answer key (conflict)	$+0.4$ [-1.2, $+2.2$]	1.000 (0.798)	n.s.	yes
H23	CB majority, gpt-6.1-terra: dir. flag vs answer key (conflict)	$+2.3$ [$+0.3$, $+4.7$]	0.515 (0.114)	n.s.	yes
H25: Claude					
H25	ConflictBank, claude-haiku-5-5: neutral flag vs plain (conflict)	$+0.7$ [-2.2, $+3.4$]	1.000 (0.791)	n.s.	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+22.7$ [$+18.6$, $+26.9$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > one-wrong flag (conflict)	$+16.8$ [$+13.0$, $+20.8$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-haiku-5-5: LLM dir. flag > plain (all q.)	-4.3 [-8.1, -0.7]	0.145 (0.042)	not supported	yes
H25	ConflictBank, claude-sonnet-5-5: neutral flag vs plain (conflict)	$+3.0$ [-0.3, $+6.3$]	0.634 (0.192)	n.s.	yes
H25	ConflictBank, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+13.3$ [$+9.7$, $+17.3$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-sonnet-5-5: dir. flag > one-wrong flag (conflict)	$+12.3$ [$+8.7$, $+16.3$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-sonnet-5-5: LLM dir. flag > plain (all q.)	-2.2 [-5.8, $+1.0$]	0.993 (0.344)	not supported	yes
H25	RAMDocs, claude-haiku-5-5: neutral flag vs plain (conflict)	$+1.2$ [-2.6, $+5.3$]	1.000 (0.732)	n.s.	yes
H25	RAMDocs, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+27.1$ [$+20.1$, $+34.1$]	<0.001 (<0.001)	supported	yes
H25	RAMDocs, claude-sonnet-5-5: neutral flag vs plain (conflict)	0.0 [-4.3, $+4.3$]	1.000 (1.000)	n.s.	yes
H25	RAMDocs, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+14.4$ [$+7.9$, $+21.6$]	0.002 (<0.001)	supported	yes
H25	CB majority, claude-haiku-5-5: dir. flag > plain (conflict)	$+57.3$ [$+52.0$, $+62.7$]	<0.001 (<0.001)	supported	yes
H25	CB majority, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	$+6.8$ [$+2.1$, $+11.4$]	0.043 (0.012)	significant	yes
H26: RAGuard (real misleading posts)					
H26	RAGuard, gpt-4o-mini: neutral flag vs plain (conflict)	$+3.2$ [$+0.1$, $+6.3$]	0.426 (0.087)	n.s.	yes
H26	RAGuard, gpt-4o-mini: dir. flag > plain (conflict)	$+9.9$ [$+5.6$, $+14.2$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, gpt-4o-mini: dir. flag > neutral flag (conflict)	$+6.7$ [$+2.9$, $+10.5$]	0.008 (0.001)	supported	yes
H26	RAGuard, gpt-4o-mini: one-wrong flag vs plain (conflict)	-5.7 [-10.2, -1.2]	0.161 (0.030)	n.s.	yes
H26	RAGuard, gpt-4o-mini: LLM dir. flag vs plain (conflict)	$+0.6$ [-2.8, $+4.0$]	1.000 (0.879)	n.s.	yes
H26	RAGuard, gpt-6-luna: neutral flag vs plain (conflict)	0.0 [-1.7, $+1.7$]	1.000 (1.000)	n.s.	yes
H26	RAGuard, gpt-6-luna: dir. flag > plain (conflict)	$+8.8$ [$+5.7$, $+12.1$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, gpt-6-luna: dir. flag > neutral flag (conflict)	$+8.8$ [$+5.7$, $+12.2$]	<0.001 (<0.001)	supported	yes

Table 13: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 2 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H26: RAGuard (real misleading posts)					
H26	RAGuard, gpt-6-luna: one-wrong flag vs plain (conflict)	−7.0 [−10.3, −3.9]	<0.001 (<0.001)	significant	yes
H26	RAGuard, gpt-6-luna: LLM dir. flag vs plain (conflict)	+3.1 [−0.1, +6.2]	0.511 (0.117)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: neutral flag vs plain (conflict)	+2.4 [0.0, +5.1]	0.731 (0.188)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > plain (conflict)	+6.8 [+3.4, +10.5]	0.005 (<0.001)	supported	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+4.4 [+1.4, +7.8]	0.126 (0.021)	not supported	yes
H26	RAGuard, gpt-6.1-terra: one-wrong flag vs plain (conflict)	+0.7 [−3.4, +4.8]	1.000 (0.935)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	+2.0 [−1.0, +5.1]	1.000 (0.392)	n.s.	yes
H26	RAGuard, claude-haiku-5-5: neutral flag vs plain (conflict)	+0.3 [−2.0, +2.7]	1.000 (0.932)	n.s.	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > plain (conflict)	+12.7 [+9.2, +16.3]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > neutral flag (conflict)	+12.4 [+9.0, +15.9]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-haiku-5-5: one-wrong flag vs plain (conflict)	+5.9 [+2.9, +9.0]	0.005 (<0.001)	significant	yes
H26	RAGuard, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	+5.0 [+1.8, +8.2]	0.025 (0.004)	significant	yes
H26	RAGuard, claude-sonnet-5-5: neutral flag vs plain (conflict)	+4.4 [+0.7, +8.2]	0.321 (0.062)	n.s.	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > plain (conflict)	+20.4 [+15.6, +25.2]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	+16.0 [+11.6, +20.4]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-sonnet-5-5: LLM dir. flag vs plain (conflict)	+7.1 [+3.1, +11.2]	0.022 (0.004)	significant	yes
H21: risk and coverage					
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	−13.3 [−17.6, −9.1]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	−26.8 [−31.6, −22.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	−6.9 [−9.9, −4.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	−24.3 [−28.6, −20.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	−2.3 [−6.0, +1.3]	1.000 (0.397)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	−17.3 [−21.7, −13.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	+0.1 [−2.6, +2.8]	1.000 (1.000)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	−21.8 [−26.1, −17.6]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	+0.3 [−3.4, +4.2]	1.000 (0.975)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	−24.9 [−29.7, −20.1]	<0.001 (<0.001)	significant	–

Table 14: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 3 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	−3.8 [−6.8, −0.9]	0.079 (0.029)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	−25.7 [−30.0, −21.3]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	−1.3 [−5.0, +2.3]	1.000 (0.695)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	−14.3 [−18.3, −10.3]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	−4.1 [−6.9, −1.3]	0.025 (0.008)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	−20.9 [−25.1, −16.8]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+34.3 [+29.3, +39.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−4.6 [−7.2, −2.0]	0.013 (0.002)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	+17.0 [+13.6, +20.6]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	−11.1 [−14.1, −8.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	+3.0 [+0.7, +5.3]	0.281 (0.041)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	−1.3 [−2.7, −0.3]	1.000 (0.188)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	−0.3 [−0.9, +0.1]	1.000 (0.478)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.9, +0.8]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+0.3 [−0.1, +0.9]	1.000 (0.478)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.3, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, selective accuracy (ratio) (conflict)	+0.2 [−4.8, +5.2]	1.000 (0.996)	n.s.	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, selective accuracy (ratio) (conflict)	+28.2 [+22.8, +33.5]	0.002 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, selective accuracy (ratio) (conflict)	+1.2 [−2.2, +4.8]	1.000 (0.599)	n.s.	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, selective accuracy (ratio) (conflict)	+29.9 [+25.0, +34.7]	0.002 (<0.001)	significant	–

Table 15: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 4 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, selective accuracy (ratio) (conflict)	-0.1 [-4.0, $+3.7$]	1.000 (1.000)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+16.1$ [$+11.5$, $+20.6$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+0.4$ [-2.3, $+3.3$]	1.000 (0.861)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+23.5$ [$+19.2$, $+27.9$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, selective accuracy (ratio) (conflict)	0.0 [-3.9, $+3.9$]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+24.2$ [$+19.4$, $+28.9$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+1.6$ [-1.7, $+4.7$]	1.000 (0.478)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+26.4$ [$+21.9$, $+31.1$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, selective accuracy (ratio) (conflict)	-0.3 [-4.0, $+3.3$]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+14.7$ [$+10.3$, $+19.0$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+3.9$ [$+1.2$, $+6.7$]	0.046 (0.011)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+21.1$ [$+17.0$, $+25.4$]	0.002 ($<$0.001)	significant	–
H24: detector versus generator reliability (crossed)					
H24	gain rises with detector-minus-generator reliability, Spearman over 30 cells	$\rho = +0.64$	$<$0.001	supported	–
H24	per-question slope of gain on reliability gap, within cells	$+47.6$ [$+43.2$, $+51.9$]	$<$0.001	supported	–

Table 16: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 5 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	Qwen2.5-7B, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+8.7 [+2.7, +14.3]	0.093 (0.011)	n.s.	no
H27	Qwen2.5-7B, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+4.3 [−1.7, +10.3]	1.000 (0.287)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+4.3 [−1.3, +10.0]	1.000 (0.251)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	−4.0 [−9.7, +1.7]	1.000 (0.287)	n.s.	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+11.1 [+7.2, +15.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+14.6 [+9.6, +19.7]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+14.1 [+9.8, +18.6]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	−1.4 [−6.0, +3.2]	1.000 (0.682)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+10.3 [+6.6, +14.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+11.0 [+7.1, +15.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+13.0 [+8.0, +18.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+0.7 [−4.7, +6.0]	1.000 (0.967)	n.s.	yes
H27	gpt-4.1, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+9.7 [+4.7, +14.7]	0.006 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+1.7 [−3.0, +6.3]	1.000 (0.688)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+6.2 [+1.1, +11.4]	0.279 (0.038)	n.s.	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+5.8 [+0.2, +11.3]	0.485 (0.075)	n.s.	yes
H27	gpt-4o-mini, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+11.8 [+6.6, +17.0]	0.005 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+7.1 [+1.8, +12.6]	0.180 (0.022)	n.s.	yes
H27	gpt-4o-mini, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+8.1 [+2.9, +13.2]	0.037 (0.005)	significant	yes
H27	gpt-4o-mini, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+0.1 [−4.9, +5.2]	1.000 (1.000)	n.s.	yes
H27	gpt-6-luna, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+4.9 [−0.4, +10.3]	0.785 (0.137)	n.s.	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+3.9 [−0.8, +8.6]	0.992 (0.188)	n.s.	yes
H27	gpt-6-luna, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+13.7 [+8.4, +18.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+6.0 [+0.6, +11.7]	0.485 (0.069)	n.s.	yes
H27	gpt-6-luna, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+8.2 [+3.7, +12.8]	0.005 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	−5.8 [−10.6, −1.0]	0.279 (0.036)	n.s.	yes

Table 17: Pre-registered tests added in the seventh (§15; H27–H29) (part 1 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-6.1-terra, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+10.7 [+5.0, +16.0]	0.008 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	−6.3 [−12.0, −0.7]	0.485 (0.071)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+9.7 [+4.7, +14.7]	0.005 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+9.7 [+5.0, +14.3]	0.008 (<0.001)	significant	yes
H27	Qwen2.5-7B, detector gpt-4.1: hint vs plain (conflict)	+7.3 [+2.3, +12.3]	0.071 (0.011)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-4o-mini: hint vs plain (conflict)	+3.0 [−2.0, +8.0]	1.000 (0.400)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: hint vs plain (conflict)	+7.7 [+3.0, +12.7]	0.038 (0.006)	significant	no
H27	Qwen2.5-7B, detector gpt-6.1-terra: hint vs plain (conflict)	+12.0 [+7.0, +17.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: hint vs plain (conflict)	−5.4 [−8.8, −2.1]	0.032 (0.004)	significant	yes
H27	claude-haiku-5-5, detector claude-sonnet-5-5: hint vs plain (conflict)	−6.3 [−9.4, −3.2]	0.004 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4.1: hint vs plain (conflict)	−3.1 [−5.8, −0.4]	0.341 (0.054)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: hint vs plain (conflict)	−2.7 [−5.6, +0.1]	0.681 (0.120)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-6-luna: hint vs plain (conflict)	−5.3 [−8.2, −2.6]	0.011 (0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: hint vs plain (conflict)	−6.9 [−9.9, −4.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4.1: hint vs plain (conflict)	−1.3 [−3.3, +0.3]	1.000 (0.392)	n.s.	yes
H27	gpt-4.1, detector gpt-4o-mini: hint vs plain (conflict)	−3.0 [−6.7, +0.3]	1.000 (0.206)	n.s.	yes
H27	gpt-4.1, detector gpt-6-luna: hint vs plain (conflict)	−3.3 [−7.3, +0.3]	1.000 (0.199)	n.s.	yes
H27	gpt-4.1, detector gpt-6.1-terra: hint vs plain (conflict)	+3.3 [−0.7, +7.0]	1.000 (0.197)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: hint vs plain (conflict)	+6.8 [+2.8, +10.9]	0.027 (0.003)	significant	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: hint vs plain (conflict)	+10.1 [+6.4, +13.8]	<0.001 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4.1: hint vs plain (conflict)	+6.8 [+3.7, +10.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4o-mini: hint vs plain (conflict)	+1.3 [−0.4, +3.2]	1.000 (0.293)	n.s.	yes
H27	gpt-4o-mini, detector gpt-6-luna: hint vs plain (conflict)	+5.6 [+2.2, +9.0]	0.036 (0.005)	significant	no
H27	gpt-4o-mini, detector gpt-6.1-terra: hint vs plain (conflict)	+9.4 [+5.9, +13.1]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-haiku-5-5: hint vs plain (conflict)	+9.4 [+5.7, +13.2]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: hint vs plain (conflict)	+14.2 [+10.4, +18.2]	<0.001 (<0.001)	significant	yes

Table 18: Pre-registered tests added in the seventh (§15; H27–H29) (part 2 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-6-luna, detector gpt-4.1: hint vs plain (conflict)	+7.7 [+4.2, +11.2]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: hint vs plain (conflict)	+2.6 [−1.3, +6.4]	1.000 (0.325)	n.s.	yes
H27	gpt-6-luna, detector gpt-6-luna: hint vs plain (conflict)	+2.9 [+0.2, +5.6]	0.462 (0.077)	n.s.	yes
H27	gpt-6-luna, detector gpt-6.1-terra: hint vs plain (conflict)	+13.1 [+9.4, +16.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4.1: hint vs plain (conflict)	−3.3 [−7.7, +1.0]	1.000 (0.241)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: hint vs plain (conflict)	−5.3 [−9.3, −1.3]	0.194 (0.030)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-6-luna: hint vs plain (conflict)	−7.0 [−11.0, −3.3]	0.006 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: hint vs plain (conflict)	−7.0 [−11.0, −3.0]	0.027 (0.003)	significant	yes
H27	Qwen2.5-7B, detector gpt-4.1: cell vs detector model alone (conflict)	+1.7 [−4.0, +7.3]	1.000 (0.740)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-4o-mini: cell vs detector model alone (conflict)	+6.6 [+0.8, +12.3]	0.312 (0.054)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: cell vs detector model alone (conflict)	+8.8 [+3.6, +14.0]	0.021 (0.004)	significant	yes
H27	Qwen2.5-7B, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−14.7 [−20.3, −9.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: cell vs detector model alone (conflict)	+5.7 [+2.1, +9.3]	0.027 (0.005)	significant	no
H27	claude-haiku-5-5, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−0.7 [−4.9, +3.6]	1.000 (0.879)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-4.1: cell vs detector model alone (conflict)	+13.1 [+8.1, +18.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.6 [+5.4, +17.4]	0.009 (0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6-luna: cell vs detector model alone (conflict)	+18.2 [+13.3, +23.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−2.1 [−7.6, +3.3]	1.000 (0.585)	n.s.	yes
H27	gpt-4.1, detector gpt-4.1: cell vs detector model alone (conflict)	+11.7 [+6.7, +16.7]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.2 [+5.2, +17.1]	0.006 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6-luna: cell vs detector model alone (conflict)	+17.4 [+12.4, +22.3]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−3.3 [−8.7, +2.0]	1.000 (0.377)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: cell vs detector model alone (conflict)	−2.7 [−7.7, +2.3]	1.000 (0.432)	n.s.	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−8.7 [−13.7, −3.7]	0.018 (0.003)	significant	no
H27	gpt-4o-mini, detector gpt-4.1: cell vs detector model alone (conflict)	+5.0 [−0.2, +10.1]	0.671 (0.119)	n.s.	yes
H27	gpt-4o-mini, detector gpt-4o-mini: cell vs detector model alone (conflict)	+8.4 [+3.2, +13.8]	0.021 (0.003)	significant	yes

Table 19: Pre-registered tests added in the seventh (§15; H27–H29) (part 3 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-4o-mini, detector gpt-6-luna: cell vs detector model alone (conflict)	+11.2 [+5.8, +16.7]	0.004 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−12.3 [−18.0, −6.7]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-haiku-5-5: cell vs detector model alone (conflict)	+1.1 [−3.8, +5.8]	1.000 (0.785)	n.s.	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−4.0 [−8.3, +0.3]	0.671 (0.130)	n.s.	yes
H27	gpt-6-luna, detector gpt-4.1: cell vs detector model alone (conflict)	+10.2 [+4.8, +15.6]	0.009 (0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.0 [+4.9, +17.0]	0.011 (0.002)	significant	yes
H27	gpt-6-luna, detector gpt-6-luna: cell vs detector model alone (conflict)	+11.1 [+7.3, +15.0]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−12.1 [−17.2, −6.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4.1: cell vs detector model alone (conflict)	+15.7 [+10.7, +20.7]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: cell vs detector model alone (conflict)	+10.2 [+4.2, +16.2]	0.018 (0.003)	significant	yes
H27	gpt-6.1-terra, detector gpt-6-luna: cell vs detector model alone (conflict)	+22.1 [+17.2, +27.0]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: cell vs detector model alone (conflict)	+2.7 [−0.3, +5.7]	0.925 (0.199)	n.s.	yes
H28: LLM flags on genuine conflicts					
H28	WikiContradict, gpt-4o-mini: LLM dir. flag vs plain	−6.5 [−10.3, −2.8]	0.001 (0.002)	significant	–
H28	WikiContradict, gpt-4o-mini: LLM removal vs plain	−9.1 [−13.3, −5.1]	<0.001 (<0.001)	significant	–
H28	WikiContradict, gpt-6-luna: LLM dir. flag vs plain	−29.0 [−34.5, −23.6]	<0.001 (<0.001)	significant	–
H28	WikiContradict, gpt-6-luna: LLM removal vs plain	−30.0 [−35.6, −24.5]	<0.001 (<0.001)	significant	–
H28	WikiContradict, claude-haiku-5-5: LLM dir. flag vs plain	−27.7 [−33.2, −22.3]	<0.001 (<0.001)	significant	–
H28	WikiContradict, claude-haiku-5-5: LLM removal vs plain	−28.2 [−33.6, −22.9]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM dir. flag vs plain	−4.0 [−8.4, 0.0]	0.100 (0.156)	n.s.	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM removal vs plain	−21.2 [−29.6, −13.1]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM dir. flag vs plain, gold recall	−3.6 [−6.8, −0.7]	0.024 (0.042)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM removal vs plain, gold recall	−17.4 [−22.4, −12.4]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM dir. flag vs plain	−14.8 [−21.5, −8.8]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM removal vs plain	−29.3 [−38.0, −20.9]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM dir. flag vs plain, gold recall	−8.9 [−12.4, −5.7]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM removal vs plain, gold recall	−20.3 [−25.0, −15.7]	<0.001 (<0.001)	significant	–

Table 20: Pre-registered tests added in the seventh (§15; H27–H29) (part 4 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H28: LLM flags on genuine conflicts					
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM dir. flag vs plain	-5.1 [-10.1, -0.3]	0.068 (0.109)	n.s.	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM removal vs plain	-25.6 [-34.0, -17.8]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM dir. flag vs plain, gold recall	-2.7 [-5.8, $+0.2$]	0.089 (0.139)	n.s.	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM removal vs plain, gold recall	-16.6 [-21.2, -12.0]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM dir. flag vs plain	-1.0 [-6.1, $+3.0$]	1.000 (1.000)	n.s.	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM removal vs plain	-27.3 [-36.4, -18.2]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM dir. flag vs plain, gold recall	0.0 [-2.4, $+2.2$]	1.000 (1.000)	n.s.	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM removal vs plain, gold recall	-18.5 [-23.6, -13.5]	<0.001 (<0.001)	significant	–
H29: artefact probe of detector direction					
H29	gpt-4o-mini detector, reversed order: direction accuracy vs original	-1.3 [-7.3, $+4.7$]	1.000 (0.811)	n.s.	–
H29	gpt-4o-mini detector, paraphrased: direction accuracy vs original	$+1.0$ [-2.7, $+4.7$]	1.000 (0.806)	n.s.	–
H29	gpt-6-luna detector, reversed order: direction accuracy vs original	-1.7 [-6.0, $+2.7$]	1.000 (0.644)	n.s.	–
H29	gpt-6-luna detector, paraphrased: direction accuracy vs original	-1.3 [-5.7, $+3.0$]	1.000 (0.747)	n.s.	–
H29	gpt-4o-mini detector, reversed order: gpt-4o-mini's gain vs original	-1.9 [-8.4, $+4.7$]	1.000 (0.699)	n.s.	–
H29	gpt-4o-mini detector, paraphrased: gpt-4o-mini's gain vs original	$+1.7$ [-3.1, $+6.3$]	1.000 (0.627)	n.s.	–
H29	gpt-6-luna detector, reversed order: gpt-4o-mini's gain vs original	-4.6 [-8.9, -0.2]	0.196 (0.081)	n.s.	–
H29	gpt-6-luna detector, paraphrased: gpt-4o-mini's gain vs original	-1.2 [-5.1, $+2.7$]	1.000 (0.680)	n.s.	–

Table 21: Pre-registered tests added in the seventh (§15; H27–H29) (part 5 of 5), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H30: the detector's verdict as an answer key					
H30	Qwen2.5-7B, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+3.0 [−2.0, +8.0]	1.000 (0.397)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+3.0 [−1.7, +7.7]	1.000 (0.353)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+2.7 [−1.3, +7.0]	1.000 (0.377)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+3.3 [+0.3, +6.3]	1.000 (0.082)	n.s.	yes
H30	claude-haiku-5-5, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	+0.9 [−1.3, +3.2]	1.000 (0.605)	n.s.	yes
H30	claude-haiku-5-5, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	+1.2 [−1.4, +3.9]	1.000 (0.503)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+1.8 [−1.0, +4.6]	1.000 (0.331)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+1.0 [−1.3, +3.4]	1.000 (0.591)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+2.1 [−0.4, +4.8]	1.000 (0.208)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+2.7 [+0.6, +4.9]	0.425 (0.035)	n.s.	yes
H30	gpt-4o-mini, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	+1.4 [−2.0, +4.9]	1.000 (0.548)	n.s.	yes
H30	gpt-4o-mini, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	−2.0 [−6.3, +2.2]	1.000 (0.486)	n.s.	yes
H30	gpt-4o-mini, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+2.1 [−1.7, +5.8]	1.000 (0.397)	n.s.	yes
H30	gpt-4o-mini, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+2.4 [−0.8, +5.8]	1.000 (0.241)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+1.4 [−1.9, +4.8]	1.000 (0.540)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+2.2 [−0.9, +5.3]	1.000 (0.291)	n.s.	yes
H30	gpt-6-luna, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	−0.8 [−3.0, +1.4]	1.000 (0.654)	n.s.	yes
H30	gpt-6-luna, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	−1.2 [−3.1, +0.7]	1.000 (0.352)	n.s.	yes
H30	gpt-6-luna, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	−1.9 [−3.8, −0.2]	1.000 (0.088)	n.s.	yes
H30	gpt-6-luna, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+0.2 [−1.0, +1.7]	1.000 (0.937)	n.s.	yes
H30	gpt-6-luna, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	−0.7 [−2.7, +1.4]	1.000 (0.699)	n.s.	yes
H30	gpt-6-luna, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+0.3 [−2.0, +2.8]	1.000 (0.934)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4.1: verdict key vs hint (conflict)	+5.7 [+0.3, +10.7]	0.393 (0.073)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4o-mini: verdict key vs hint (conflict)	+1.3 [−4.3, +6.7]	1.000 (0.811)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6-luna: verdict key vs hint (conflict)	+1.7 [−3.7, +6.7]	1.000 (0.709)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6.1-terra: verdict key vs hint (conflict)	−7.3 [−12.7, −2.3]	0.095 (0.015)	n.s.	yes

Table 22: Pre-registered tests added in the eighth (§16; H30–H31) (part 1 of 3), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H30: the detector's verdict as an answer key					
H30	claude-haiku-5-5, detector claude-haiku-5-5: verdict key vs hint (conflict)	+10.2 [+6.3, +14.2]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+13.3 [+8.8, +17.9]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-4.1: verdict key vs hint (conflict)	+12.3 [+8.6, +16.2]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-4o-mini: verdict key vs hint (conflict)	−2.4 [−6.8, +1.9]	1.000 (0.400)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6-luna: verdict key vs hint (conflict)	+8.2 [+4.3, +12.1]	0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-6.1-terra: verdict key vs hint (conflict)	+8.3 [+4.6, +12.2]	<0.001 (<0.001)	significant	yes
H30	gpt-4o-mini, detector claude-haiku-5-5: verdict key vs hint (conflict)	+4.8 [+0.1, +9.3]	0.393 (0.082)	n.s.	yes
H30	gpt-4o-mini, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+7.8 [+3.2, +12.3]	0.025 (0.004)	significant	yes
H30	gpt-4o-mini, detector gpt-4.1: verdict key vs hint (conflict)	+9.7 [+5.1, +14.2]	0.005 (<0.001)	significant	yes
H30	gpt-4o-mini, detector gpt-4o-mini: verdict key vs hint (conflict)	+4.7 [−0.7, +9.9]	0.440 (0.138)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6-luna: verdict key vs hint (conflict)	+6.7 [+2.0, +11.3]	0.075 (0.011)	n.s.	no
H30	gpt-4o-mini, detector gpt-6.1-terra: verdict key vs hint (conflict)	−2.1 [−6.7, +2.4]	1.000 (0.501)	n.s.	yes
H30	gpt-6-luna, detector claude-haiku-5-5: verdict key vs hint (conflict)	+5.7 [+0.4, +10.9]	0.388 (0.067)	n.s.	yes
H30	gpt-6-luna, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+5.1 [+0.2, +10.0]	0.393 (0.080)	n.s.	yes
H30	gpt-6-luna, detector gpt-4.1: verdict key vs hint (conflict)	+15.6 [+10.8, +20.3]	<0.001 (<0.001)	significant	yes
H30	gpt-6-luna, detector gpt-4o-mini: verdict key vs hint (conflict)	+5.8 [+0.2, +11.4]	0.393 (0.083)	n.s.	yes
H30	gpt-6-luna, detector gpt-6-luna: verdict key vs hint (conflict)	+8.9 [+4.1, +13.7]	0.006 (<0.001)	significant	yes
H30	gpt-6-luna, detector gpt-6.1-terra: verdict key vs hint (conflict)	−6.1 [−11.0, −1.2]	0.209 (0.034)	n.s.	yes
H31: randomised direction					
H31	gpt-4o-mini: right-flag effect (conflict)	+29.4 [+22.6, +36.3]	<0.001 (<0.001)	supported	–
H31	gpt-6-luna: right-flag effect (conflict)	+33.5 [+26.8, +40.3]	<0.001 (<0.001)	supported	–
H31	claude-haiku-5-5: right-flag effect (conflict)	+24.1 [+18.0, +30.2]	<0.001 (<0.001)	supported	–
H31	gpt-6.1-terra: right-flag effect (conflict)	+17.6 [+11.3, +23.9]	<0.001 (<0.001)	supported	–
H31	gpt-4o-mini: wrong-flag effect (conflict)	−31.2 [−39.0, −23.4]	<0.001 (<0.001)	supported	–
H31	gpt-6-luna: wrong-flag effect (conflict)	−44.9 [−52.0, −37.8]	<0.001 (<0.001)	supported	–
H31	claude-haiku-5-5: wrong-flag effect (conflict)	−49.2 [−56.5, −42.1]	<0.001 (<0.001)	supported	–
H31	gpt-6.1-terra: wrong-flag effect (conflict)	−64.5 [−72.3, −56.0]	<0.001 (<0.001)	supported	–

Table 23: Pre-registered tests added in the eighth (§16; H30–H31) (part 2 of 3), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

racy exceeds plain accuracy (gpt-4o-mini 76.1% against 60.5%). The gpt-4o-mini detector names the false page as suspect on 36% of items and the correct page on 48%, so the frozen account predicts harm; observed effects are smaller (Table 44). Auditing the labels (247 items), the judge finds the correct page supporting the gold answer for 68%, the misinformation page the wrong answer for 55%, and both for 37%. Long, noisy pages leave the judge unsure of many stances, so the dataset's oracle, like RAGuard's, is a label transfer.

F Reproducibility

Runs. One script produces every number from the committed per-call responses, token counts, detector outputs and configurations. The pre-registration was committed before any confirmatory run; deviations follow.

Deviations from the pre-registration.

1. *Primary metric.* The first draft named the bidirectional metric primary. Before the confirmatory run we switched to one-way matching, because it is closer to MADAM-RAG's published score on the pilot split (Section 4); bidirectional became the robustness metric. Every table reports both.

2. *Second generator.* The plan named an open-weights model. None was available before the confirmatory run, so we used gpt-4.1 (H9). We later added Qwen2.5-7B-Instruct, registered before its run. At about 3 s per local call, its runs use one seed, a change made before inspecting any Qwen result.

3. *CONFLICTS judge.* The plan called for an outside-family judge validated on about 100 human labels. Neither was available, so CONFLICTS is scored, as exploratory, only on items with a reference answer (three seeds).

4. *H4.* The pre-registration counted 54 false-flag cells. On clean questions, cells that differ only in direction accuracy render identical prompts (their scores differ by exactly 0), so they are one comparison: 18 in all, 9 per format. We report H4a per format. The cell-permutation p-value for the dose-response correlation assumed exchangeable cells, but they are nested, so we use a question-level bootstrap CI. H4c had no threshold, so we report a count.

5. *H5.* A non-significant difference is not equivalence. We add a TOST with a ± 3-point margin, chosen in the addendum after the confirmatory run, so it is post hoc.

6. *H6.* The first draft's test (grid prediction inside the actual effect's CI) was lenient. We now report the pre-registered test (actual effect inside a bootstrap prediction interval) and a paired test of the two spaces' errors.

7. *Placement modes.* For H6, detectors without direction ran under both `remove_suspect` and `remove`; as the labels render identical prompts, the response cache served every duplicate.

8. *Significance.* Earlier drafts corrected different tables differently; one scheme now applies throughout (Section 4), so some exploratory differences once bold are not significant.

9. *Single- and multi-answer split.* The split of conflict questions by number of gold answers (Appendix C.1) was chosen after seeing the minimal-skeleton run and is exploratory.

10. *Second registration.* We later registered H12–H14 (ConflictBank, the neutral flag with a resolution instruction, and the published baselines), the Qwen2.5-7B detector and the ConflictBank sampling rule. Its gpt-4.1 runs use seed 0 only, to save budget. One ConflictBank gpt-4o-mini run was resumed from the response cache after a transient API error.

11. *Third and fourth registrations.* H15–H18 (§11) and H19 (§12) were committed before their runs. To save budget, §12 dropped two §11 runs before they started: gpt-4.1 under majority misinformation (replaced by gpt-6.1-terra in H17) and gpt-4.1 seeds 1–2, so H14 with gpt-4.1 stays on seed 0 as first registered, and every gpt-4.1 baseline comparison is seed-0 matched.

12. *Reasoning models.* gpt-6-luna and gpt-6.1-terra accept only their default temperature and require `max_completion_tokens`; we used their defaults with a 4,096-token budget. They returned 25 empty responses (almost all closed-book) because reasoning used the whole budget; we score them wrong.

13. *Fifth and sixth registrations.* H20–H24 (§13) and H25–H26 with Claude and RAGuard (§14) were committed before their runs. The planned `cross_mj` runs completed within budget. Gemini, planned as a generator, was not run: no

API key was available then. On our endpoint, Claude rejects a temperature and JSON mode; it ran at its default, and as a detector its JSON goes to the regex fallback. The answer parser unwraps paired tags (`<answer>X</answer>`), a format no earlier result contained.

14. *Permutation resolution.* With 10,000 permutations the smallest p-value is 10^{-4}, which cannot survive Holm over 636 exploratory comparisons. After the §14 runs, before writing them up, we extended to 200,000 permutations whenever none of the first 10,000 reached the observed statistic. Only p-values at their floor change, and only downward.

15. *H24 additions.* The cross-cell break-even point, the count of positive cells and the mechanism score (credit for naming the false side where the generator errs, charge for marking a correct passage where it is right) were defined after seeing the H24 cells and are exploratory.

16. *Seventh registration.* H27–H29 and the reanalyses of §15 were committed before their runs. To finish within the session, the §15 runs ran in parallel tracks, not strictly in priority order; spend never approached the $195 limit, so no run was dropped. The question-level model adds statsmodels 0.14.5. H21(c) is now a ratio, correct over answered. The registered per-question mean averaged the two conditions over different questions (those not abstained in every seed), so an earlier draft's selective accuracies did not match Table 2. Both measures are in the released statistics.

17. *Eighth registration.* H30–H31 and the reanalyses of §16 were committed before their runs, which ran in parallel tracks within budget. The pooled estimate now uses a multilevel model, not the DerSimonian–Laird bootstrap. The moderators were recoded (answer policy; natural misinformation is RAGuard only). The utility analysis uses the pooled estimate's settings. "Following" a flag now has one definition for right and wrong flags (§16.3); an earlier draft compared correct answers with any wrong answer.

18. *Ninth registration.* The re-analyses, corrected baselines and semantic re-score of §17 were committed before their runs, as were its two deviations (§17.6). The judge is gemini-3.5-flash-lite, because the named Gemini 2.5 Flash was closed to new users. The registered 1,500-character NLI window flags no ConflictBank pair, so we add a 300-character window, chosen after seeing detection output only, and report the registered window's results unchanged.

19. *Tenth and eleventh registrations.* The §18 pair-rule counts were computed during review triage, before §18 was written (§18.5); its other analyses were not. §19 (H32–H38) records no deviation.

20. *Run interruptions.* Several runs were stopped and resumed (a transient API error; a reordered run queue), keeping completed rows and re-requesting only missing ones; no row was re-sampled.

The addenda (H10–H38 and the re-analyses above) and the Qwen seed change were each committed before their runs. The commit history is the only timestamp; we used no third-party registry (van Miltenburg et al., 2021).

RAMDocs document types are imperfect. In the release, 163 documents typed *correct* contain a wrong answer, some as injected misinformation, others incidentally. Pairs touching them are excluded from pair-level metrics. Questions whose only sign of conflict is such a document (55 of 500; 35 of the confirmatory split) count only in "All". Treating them as misinformation is a robustness check.

Power. With 139 RAMDocs conflict questions, the minimum detectable effect at 80% power is 4–12 points (median 5.8; Card et al., 2020).

Metric choice. With the one-way metric, our port of MADAM-RAG scores 28.5% on the pilot split and 32.8% on all 500 questions (published: 28.00, on an undisclosed subset); the bidirectional metric gives 38.4% on all 500. With the subset and scorer unknown, we do not claim to reproduce the published number. Normalisation lower-cases and removes punctuation and articles.

WikiContradict and CONFLICTS. A WikiContradict answer is correct if both dataset answer strings (one per passage) match a final answer under the bidirectional rule. From CONFLICTS we use the 237 questions with a reference answer (outdated-information, misinformation and no-conflict items); an answer is correct if it contains the reference answer.

Baselines. We port MADAM-RAG unchanged to our client: three debate rounds, temperature 0, 128 tokens per call. Astute RAG has no official code; we use its appendix prompts verbatim with its defaults (t=1 consolidation round, m=1 generated passage, temperature 0, 1,024 output tokens). We drop a generated passage that says "I don't know"; the answer is the text in its `<ANSWER>` tag. We port TrustRAG from its repository (commit `11dcea0`, MIT licence): SimCSE CLS embeddings (`sup-simcse-bert-base-uncased`), a ROUGE-L gate of 0.25, 2-means clustering and a cosine threshold of 0.88, then its three prompts verbatim at temperature 0.1. The registered runs omit its system message (our client then sent only user messages); the §17 rerun restores it (Appendix B). The *multi* variants change one sentence of the final step to ask for every valid answer (Astute RAG) or append our final-answer line (TrustRAG); we also report the unmodified variants.

ConflictBank. We use the QA file of the Hugging Face release (revision `92866a0`). We sample questions round-robin over relations with seed 0; distractors are other subjects' correct passages. The sampling script, item ids and seed are released; we do not redistribute the data, because the release states a licence in its README but ships no licence file.

RAGuard. We use the Hugging Face release `UCSC-IRKM/RAGuard` (revision `dae344a2`, MIT licence on the card). A post is usable if, without the "[Link Post]" placeholder, it has at least 15 words and is not a duplicate within its claim; posts are cut to 300 words and shuffled by claim id. Each supporting × misleading pair is a contradiction, with the misleading post as the misinformation side. The build script and item ids are released.

Model configuration. Models were called by these names between 2026-10-26 and 2026-10-29; served snapshot identifiers were not recorded. gpt-4o-mini and gpt-4.1 ran at temperature 0.1 with at most 256 output tokens. gpt-6-luna and gpt-6.1-terra ran at default temperature and reasoning effort, with 4,096 completion tokens including reasoning. Claude Haiku 5.5 and Sonnet 5.5 ran through Anthropic's OpenAI-compatible endpoint at their default temperature with 4,096 output tokens. Qwen2.5-7B-Instruct ran through Ollama 0.40.11 (`qwen2.5:7b-instruct`, image `845dbda0ea48`, Q4_K_M quantisation, 8,192-token context, temperature 0.1) on an Apple M4 Pro with 48 GB. Across settings and conditions, at most 2.1% of responses are empty or aborted; the share with no final-answer line has median 0.0% and maximum 20.0% (gpt-4.1, `oracle+flag-instruct-v2`). Per-condition counts are in the released statistics.

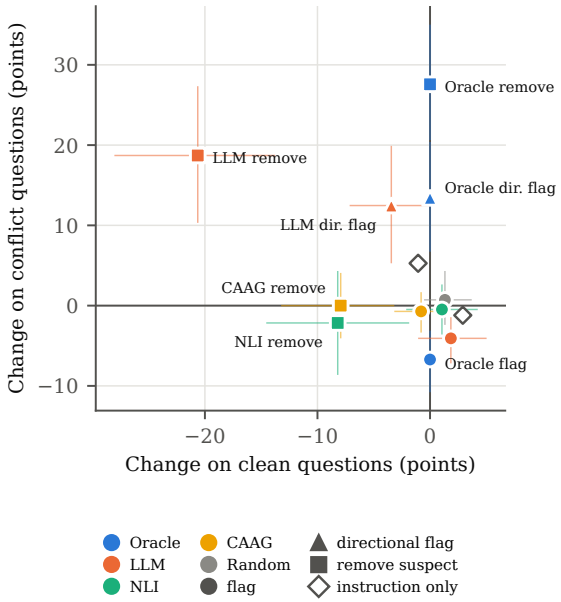


Figure 4: Change in one-way strict accuracy on RAMDocs conflict questions (y) against clean questions (x), relative to no flags (gpt-4o-mini, default skeleton, confirmatory split, 95% CIs). Colour: detector; shape: format. The oracle sits on the y-axis because it never flags a clean question. Without direction, removal deletes the second passage of each pair.

Cost and compute. All API calls, pilots included, cost $204.18. Prices per million input, cached-input and output tokens, checked on 2026-10-27: gpt-4o-mini $0.15, $0.075 and $0.60; gpt-4.1 $2.00, $0.50 and $8.00; gpt-6-luna $0.10, $0.01 and $0.50; gpt-6.1-terra $1.00, $0.05 and $5.00; Claude Haiku 5.5 $0.10, $0.01 and $0.50; Claude Sonnet 5.5 $2.00, $0.10 and $10.00. The §17.3 judge, gemini-3.5-flash-lite, cost $0.30 input and $2.50 output (checked on 2026-10-29). CAAG and NLI ran on a laptop CPU (0.2 s and 1.2 s per question).

G Synthetic Detectors and the Grid

Construction. We build each synthetic detector (Section 3.3) from the gold labels. Each gold contradiction pair is flagged with probability r. False flags are drawn from non-contradicting pairs at a

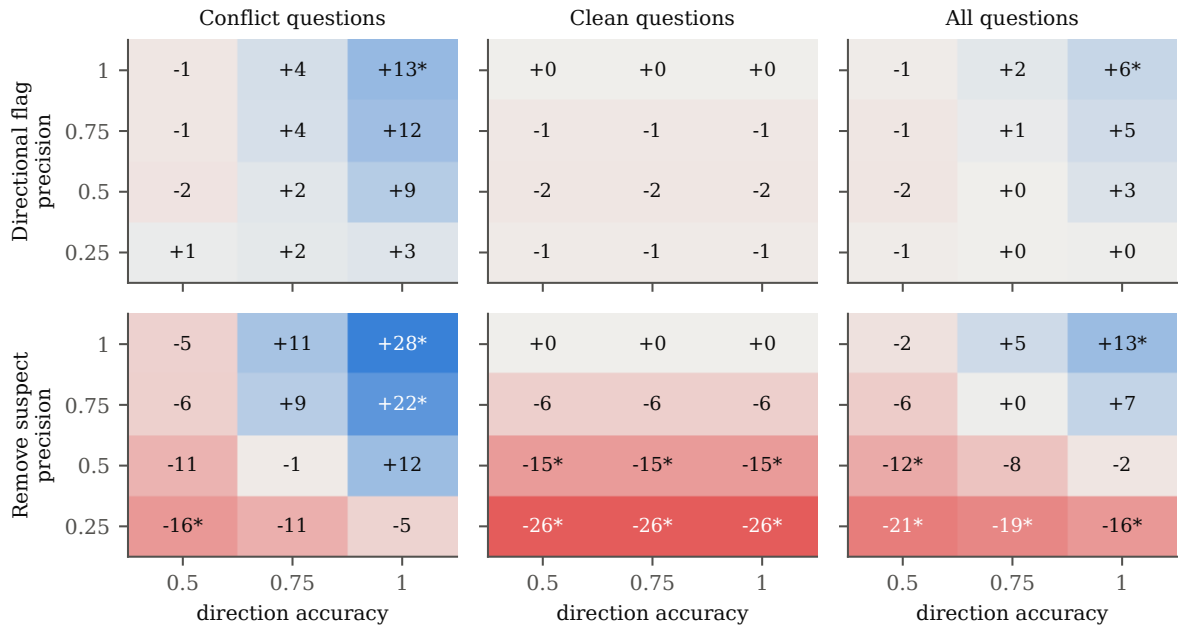


Figure 5: Synthetic detectors at recall 1: change in one-way strict accuracy (points) against no flags, by pair precision (rows) and direction accuracy (columns), for the directional flag (top) and removal (bottom). *: significant after Holm correction over the 36 cells of that format and subset. Clean questions only see false flags, so their rows do not vary with direction accuracy.

single per-pair rate, set over the whole split so that expected precision is p; clean questions therefore receive false flags too. A true flag names the gold misinformation side with probability d, and otherwise the other side. A false flag has no misinformation side, so its suspect is picked uniformly. Each pair gets one uniform draw for flagging and one for direction, and every cell reuses them. Cells are therefore nested: raising r or d, or lowering p, only adds flags or corrections, which keeps comparisons across cells paired.

Gains rise with direction accuracy; precision matters mainly for removal. Figure 5 replaces the oracle with synthetic detectors of known quality. Gains rise with direction accuracy in both formats. Only the cells with perfect direction are significant on all questions (directional flag +6, removal +13 points at perfect precision). Neighbouring cells at direction accuracy 0.75 have similar point estimates on conflict questions, so the pattern is a dose-response, not a threshold. With 36 cells under Holm, the grid has little power to separate adjacent levels.

Precision matters mainly for removal. On clean questions a cell's prompt does not depend on direction accuracy, so the false-flag cells reduce to 9 distinct comparisons per format. Removal is significantly harmful in 6 of them, the directional flag in 0 (H4a). Under removal, false flags delete correct passages, which costs clean questions 15 points at precision 0.5 and 26 at 0.25, while synthetic false directional flags cost at most two points. The harm tracks the share of clean questions that lose a correct document (Spearman ρ =-0.39 over 18 distinct cells, 95% CI [−0.74, −0.12] from a question-level bootstrap; H4b). Removal hurts clean questions significantly more than the directional flag in 5 of 9 matched cells (H4c; we set no threshold, so we report the count). Removal is both more powerful and more fragile: its best cell gains twice the directional flag's on all questions, but below precision 0.75 it is the only format that does net harm.

Placing real detectors on the grid. We compare two ways of summarising a real detector. *Pair-level*: its pair precision, recall and direction accuracy. *Document-level*: what it does to each subset's documents in a given format. For conflict questions, that is the share whose misinformation documents are all marked and the share that have a correct document marked; for clean questions, only the latter. Here "marked" means removed (remove) or headed as misinformation (directional flag). The prediction for a subset is the inverse-

distance-weighted mean effect of the four nearest grid cells in that space.

Both summaries place most real effects inside the prediction interval. Figure 2 (Appendix B) tests whether the grid predicts real detectors' effects (H6). As pre-registered, we bootstrap the whole prediction (cell effects, every detector's location and the interpolation) over questions, and ask whether the actual effect falls inside the 95% interval. Document-level locations put 7 of 8 effects inside, and pair-level locations 6. Document-level errors are smaller on average (4.1 vs. 5.3 points), but the paired difference, -1.2 [-3.6, $+0.4$] (exact sign-flip $p = 0.41$), is not significant over 8 cases. Both spaces miss the same case, the LLM detector's removal gain on conflict questions: one correctly directed pair suffices to remove a misinformation document, so the detector removes the misinformation on 61.2% of conflict questions despite its low pair recall. We chose the document-level axes after seeing the LLM detector on the pilot split. The grid is a useful first estimate of a detector's effect, but with three detectors we cannot claim that one summary predicts better than the other.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H31: randomised direction					
H31	gpt-4o-mini: right-flag effect + wrong-flag effect (conflict)	-1.9 [-12.2, $+8.6$]	0.733 (0.815)	n.s.	–
H31	gpt-6-luna: right-flag effect + wrong-flag effect (conflict)	-11.4 [-21.0, -1.8]	0.049 (0.043)	significant	–
H31	claude-haiku-5-5: right-flag effect + wrong-flag effect (conflict)	-25.1 [-34.3, -15.8]	$<$0.001 ($<$0.001)	significant	–
H31	gpt-6.1-terra: right-flag effect + wrong-flag effect (conflict)	-46.9 [-57.1, -36.4]	$<$0.001 ($<$0.001)	significant	–

Table 24: Pre-registered tests added in the eighth (§16; H30–H31) (part 3 of 3), each committed before the runs it covers. Same conventions as Table 7. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Effect (points)			Follow rate (%)		Share of headroom (%)		Net at direction precision		
Generator	Right	Wrong	Break-even [95% CI]	Right	Wrong	Right	Wrong	0.50	68.8%	82.6%
gpt-4o-mini	$+29.4$	-31.2	0.52 [0.43, 0.60]	57 [46, 67]	51 [40, 62]	59	59	-0.9	$+10.5$	$+18.8$
gpt-6-luna	$+33.5$	-44.9	0.57 [0.51, 0.64]	72 [60, 83]	77 [67, 86]	76	87	-5.7	$+9.1$	$+19.9$
Claude Haiku	$+24.1$	-49.2	0.67 [0.61, 0.74]	63 [49, 76]	64 [55, 73]	70	72	-12.5	$+1.3$	$+11.4$
gpt-6.1-terra	$+17.6$	-64.5	0.79 [0.72, 0.85]	67 [53, 80]	95 [89, 99]	62	96	-23.5	-8.0	$+3.4$

Table 25: Randomised direction (H31; ConflictBank, one-answer prompt, conflict questions), re-analysed (exploratory). *Break-even*: the direction precision at which a flag on the true pair neither helps nor hurts, $-W/(R-W)$ for right-flag effect R and wrong-flag effect W; bootstrap CI within arms. *Follow rate*: a right flag is followed when an answer that was wrong without the flag becomes correct, a wrong flag when a correct answer comes to contain the misinformation; question-bootstrap CIs. *Share of headroom*: R over the no-flag error rate of the right arm, and $-W$ over the no-flag accuracy of the wrong arm. *Net*: $pR + (1-p)W$ at direction precision p: 0.50, gpt-4o-mini's on ConflictBank, and the most precise detector's (gpt-6.1-terra). These nets assume the detector flags the true pair; real detectors also flag other pairs and clean questions. Figure 1 (right) uses the follow rates to predict the 30 crossed cells.

Setting	Conflict	Clean	5%	20%	Break-even
CB multi-answer, Qwen2.5-7B	+6.7	−1.0	−0.6	+0.5	13
CB multi-answer, gpt-4.1	+17.0	0.0	+0.9	+3.4	–
CB multi-answer, gpt-4o-mini	+20.0	−1.0	+0.1	+3.2	5
CB forced, Claude Haiku	−4.7	−1.3	−1.5	−2.0	–
CB forced, gpt-4o-mini	+4.0	−2.7	−2.3	−1.3	40
CB forced, gpt-6-luna	−2.4	−1.3	−1.4	−1.6	–
CB forced, gpt-6.1-terra	−14.3	0.0	−0.7	−2.9	–
CB majority, Claude Haiku	+6.8	−5.0	−4.4	−2.6	42
CB majority, gpt-4o-mini	+5.7	−4.7	−4.2	−2.6	45
CB majority, gpt-6-luna	+7.1	−1.3	−0.9	+0.4	16
CB majority, gpt-6.1-terra	+9.0	0.0	+0.4	+1.8	–
CB one-answer, Qwen2.5-7B	+7.3	−1.0	−0.6	+0.7	12
CB one-answer, Claude Haiku	−4.1	−5.0	−5.0	−4.8	–
CB one-answer, Claude Sonnet	−2.7	−1.0	−1.1	−1.3	–
CB one-answer, gpt-4.1	−2.3	−3.0	−3.0	−2.9	–
CB one-answer, gpt-4o-mini	+8.4	−4.7	−4.0	−2.0	36
CB one-answer, gpt-6-luna	+8.6	−1.3	−0.8	+0.6	13
CB one-answer, gpt-6.1-terra	−11.7	0.0	−0.6	−2.3	–
RAMDocs, Qwen2.5-7B (multi-answer)	+2.9	−7.1	−6.6	−5.1	71
RAMDocs, Claude Haiku	+22.5	−4.5	−3.1	+0.9	17
RAMDocs, Claude Sonnet	+12.2	−3.2	−2.4	−0.1	21
RAMDocs, gpt-4.1 (multi-answer)	+8.4	0.0	+0.4	+1.7	0
RAMDocs, gpt-4o-mini (multi-answer)	+12.5	−3.4	−2.6	−0.3	22
RAMDocs, gpt-4o-mini (minimal)	+3.6	−10.3	−9.6	−7.5	74
RAMDocs, gpt-6-luna	+29.7	−11.6	−9.6	−3.4	28
RAMDocs, gpt-6.1-terra	+38.8	−0.8	+1.2	+7.1	2
RAGuard, Claude Haiku	+5.0	+1.3	+1.5	+2.1	–
RAGuard, Claude Sonnet	+7.1	+10.0	+9.9	+9.4	–
RAGuard, gpt-4o-mini	+0.6	+1.7	+1.6	+1.4	–
RAGuard, gpt-6-luna	+3.1	−1.3	−1.1	−0.5	30
RAGuard, gpt-6.1-terra	+2.0	+1.0	+1.1	+1.2	–

Table 26: The gpt-4o-mini LLM detector's directional flag: effect (points, against no flags) on conflict and clean questions, and reweighted to a 5% and a 20% share of conflict questions (exploratory). *Break-even*: the share of conflict questions (%) above which the flag helps, where it hurts on clean questions. Oracle flags never fire on clean questions, so their reweighted effect is their conflict effect times the share.

		Plain	Oracle, conflict q.			LLM, directional flag		LLM, remove suspect	
Generator	Skeleton	conflict	flag	dir. flag	remove	conflict	clean	conflict	clean
gpt-4o-mini	default	22.1	**−6.7**	**+13.4**	**+27.6**	+12.5	−3.4	+18.7	**−20.6**
gpt-4o-mini	minimal	20.4	**−5.3**	+5.0	**+18.9**	+3.6	−10.3	+11.0	**−19.6**
gpt-4.1	default	21.8	**−8.6**	**+18.2**	**+30.2**	+8.4	0.0	**+20.6**	**−24.3**
gpt-4.1	minimal	33.6	**−11.0**	**+12.7**	**+14.9**	–	–	–	–
Qwen2.5-7B	default	19.4	−5.8	+3.6	**+18.0**	+2.9	−7.1	+7.9	**−18.3**
Qwen2.5-7B	minimal	17.3	−5.8	+2.9	**+10.8**	–	–	–	–
gpt-6-luna‡	default	14.4	**−3.1**	**+39.1**	**+38.8**	**+29.7**	**−11.6**	–	–
gpt-6.1-terra‡	default	10.8	−0.7	**+39.6**	**+39.6**	**+38.8**	−0.8	–	–

Table 27: The main effects across generators and prompt skeletons (RAMDocs hold-out, 3 seeds, one-way strict accuracy). *Plain*: accuracy (%) on conflict questions with no flags. Other columns: paired difference from plain in points. *Default* skeleton tells the model to give every valid answer; *minimal* drops that policy (Appendix A). **Bold**: significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). The LLM detector is gpt-4o-mini for every generator. Qwen2.5-7B-Instruct runs locally (4-bit), seed 0. ‡: current reasoning models at their default temperature (gpt-6-luna 3 seeds, gpt-6.1-terra seed 0).

	Pair level			Clean-q.	Direc-	Removal, per question		Latency	Cost
Detector	P	R	F1	flagged	tion	misinfo gone	correct lost	(s)	($/1k q.)
CAAG (heuristic)	21.7	7.9	11.6	34.1	–	9.4	31.7	0.23	0.000
NLI cross-encoder	20.1	29.3	23.8	61.9	–	44.6	52.4	1.18	0.000
LLM (gpt-4o-mini)	37.2	20.5	26.4	76.2	88.3	61.2	65.9	0.80	0.181

Table 28: Detectors on the RAMDocs hold-out split (300 questions; percentages). *Clean-q. flagged*: share of conflict-free questions with at least one flag. *Direction*: share of correctly flagged contradictions whose suspect side is the misinformation. *Misinfo gone*: conflict questions whose misinformation documents are all removed when removing the suspect (or, without direction, the second passage). *Correct lost*: clean questions that lose at least one correct document. Latency is per question; CAAG and NLI run locally.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	35.2	37.2	25.9	49.7	911
Prompt only					
Instruction (v1)	+1.3	+0.5	−1.0	+2.1	948
Instruction (v2)	+2.2	+1.4	+3.6	−1.1	951
Oracle detector					
Oracle, flag	−3.6	−4.0	**−7.7**	0.0	980
Oracle, directional flag	+7.0	**+7.9**	**+15.1**	0.0	998
Oracle, remove suspect	**+15.4**	**+17.5**	**+33.3**	0.0	825
Real detectors (exploratory)					
Random, flag	+0.4	+0.4	−0.2	+1.1	949
CAAG, flag	−0.9	−1.4	−1.9	−0.8	942
CAAG, remove	−4.0	−4.5	−1.0	−8.5	809
NLI, flag	+0.2	0.0	−1.4	+1.6	1021
NLI, remove	−8.0	−7.9	−3.1	−13.2	592
LLM, flag	−3.4	−3.4	−7.2	+0.8	970
LLM, directional flag	+4.7	+4.5	+14.1	−6.1	983
LLM, remove suspect	−2.1	−2.6	+18.9	**−26.5**	718
Reference					
MADAM-RAG	+5.7	+5.2	+3.6	+6.9	8196

Table 29: RAMDocs hold-out, gpt-4o-mini, 3 seeds, bidirectional strict accuracy (robustness). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

	Test (labels: *contains*)	One-way [95% CI]	p_{Holm}	Bidir.	p_{Holm}
H2	oracle+flag < plain (conflict)	−4.6 [−8.1, −1.5]	0.006	−5.2	0.003
H3	oracle+flag_directional > plain (conflict)	+11.6 [+5.6, +17.3]	<0.001	+13.3	<0.001
H3	oracle+flag_directional > oracle+flag (conflict)	+16.2 [+10.8, +21.8]	<0.001	+18.5	<0.001
H3	oracle+remove_suspect > plain (conflict)	+25.8 [+19.5, +32.2]	<0.001	+30.3	<0.001
H4	harmful cells, clean q. (dir. flag; removal); ρ	0/9; 8/9; $\rho = -0.08$ [-0.56, -0.02]			

Table 30: Label robustness: pre-registered tests with every suspect document treated as misinformation (173 conflict and 127 clean hold-out questions; this rule assigns the 35 questions the main analysis excludes). The bootstrap distribution of the H4 correlation is skewed, so its point estimate lies near the end of its percentile interval.

CAAG setting	θ_s	θ_c	P	R	F1	Clean FPR
Default	0.65	0.35	21.7	7.9	11.6	34.1
Best pair F1	0.40	0.10	20.7	49.6	29.2	87.3
Best question F1	0.55	0.15	22.2	28.8	25.1	60.3

Signals (defaults)	Pair F1	Flag rate	Clean FPR
all signals	11.6	6.2	34.1
negation only	0.6	0.4	4.0
entity only	8.6	4.3	23.0
temporal only	0.0	0.0	0.0

Table 31: CAAG on the RAMDocs hold-out (%). Top: thresholds from a sweep over 121 settings ($\theta_s \in [0.40, 0.90]$, $\theta_c \in [0.10, 0.60]$). Bottom: each signal alone.

Condition	All	Lab.	Conflict	Clean
No flags (plain)	32.7	34.5	21.8	48.4
Instruction (v2)	+5.8	+5.2	+12.2	−2.6
Oracle, flag	−4.0	−4.5	**−8.6**	0.0
Oracle, directional flag	**+8.4**	**+9.6**	**+18.2**	0.0
Oracle, remove suspect	**+14.0**	**+15.8**	**+30.2**	0.0
LLM, directional flag	+5.4	+4.4	+8.4	0.0
LLM, remove suspect	+1.0	−0.8	**+20.6**	**−24.3**
NLI, remove	−7.0	−7.3	+0.2	**−15.6**

Table 33: Replication with gpt-4.1 as generator (RAMDocs hold-out, 3 seeds, one-way strict accuracy; detector fixed to gpt-4o-mini). Same conventions as Table 6.

Condition	All	Conflict	No conflict
No flags (plain)	65.8	46.9	73.6
Instruction (v2)	+0.6	+3.9	−0.8
CAAG, flag	+1.1	+2.4	+0.6
NLI, flag	−0.3	−0.5	−0.2
LLM, flag	+2.7	+5.3	+1.6
LLM, directional flag	+0.3	+1.4	−0.2

Table 32: CONFLICTS, 237 automatically scorable questions (69 conflict), gpt-4o-mini, 3 seeds; exploratory. Contains-match accuracy (%) and differences in points. These comparisons use their own Holm family (this table), more lenient than the paper's single exploratory family; bold would mark Holm-adjusted $p < 0.05$ in it, and no difference reaches it, so none is significant under either. CAAG: flags 94.2% of conflict and 92.9% of clean questions; NLI: flags 52.2% of conflict and 43.5% of clean questions; LLM: flags 98.6% of conflict and 83.9% of clean questions.

Condition	gpt-4o-mini	gpt-4.1
No flags (plain)	14.0	29.0
Instruction (v1)	**+8.7**	–
Instruction (v2)	**−6.7**	–
Oracle, flag	**+6.5**	**+19.0**
CAAG, flag	+0.7	–
NLI, flag	+3.2	–
LLM, flag	**+6.2**	–

Table 34: WikiContradict (253 genuine conflicts between two Wikipedia passages; the reference answer is both sides), 3 seeds (gpt-4.1: the oracle flag only). First row: share of answers giving both sides (%); other rows: difference in points, **bold** = significant under the paper's single scheme (H8 is a registered family; the gpt-4.1 column is exploratory).

	RAMDocs, gpt-4o-mini			RAMDocs, gpt-4.1		ConflictBank		CB majority		
Method	Lab.	Conflict	Clean	Lab.	Conflict	Conflict	Clean	Conflict	Clean	Calls
No flags (plain), accuracy	31.3	22.1	41.5	32.8†	19.4†	51.1	89.0	21.6	89.0	1
Instruction (v2) only	+2.3	+5.3	−1.1	–	+12.2	–	–	–	–	1
Oracle, flag + instr.	−1.8	−2.4	−1.1	+1.5†	+5.8†	–	–	–	–	1
Oracle, dir. flag + instr.	+6.4	+13.2	−1.1	**+14.7**†	**+30.9**†	–	–	–	–	1
MADAM-RAG	+5.7	+3.6	+7.9	–	–	–	–	–	–	9×tok.
Astute RAG	**−15.6**	−8.4	**−23.5**	–	–	−2.1	−9.7	+1.4	−9.7	2
Astute RAG (multi)	−5.9	−1.2	−11.1	+2.3†	+13.7†	–	–	–	–	2
TrustRAG	**−17.5**	−11.3	**−24.3**	–	–	−9.7	−4.0	+6.8	−4.0	3
TrustRAG (multi)	**−18.1**	−12.2	**−24.6**	−10.9†	0.0†	–	–	–	–	3
Single call (name suspects, then answer)	+6.2	+12.0	−0.3	–	–	+7.6	−1.0	+6.8	−1.0	1
LLM detector, flag + instr.	−2.0	−2.2	−1.9	–	–	–	–	–	–	1+1
LLM detector, dir. flag + instr.	−3.5	+8.4	**−16.7**	–	–	–	–	–	–	1+1
LLM detector, dir. flag	+4.9	+12.5	−3.4	+8.7†	+14.4†	+8.4	−4.7	**+5.7**	−4.7	1+1
LLM detector, remove suspect	0.0	+18.7	**−20.6**	–	**+20.6**	+7.9	−7.0	+7.0	−7.0	1+1

Table 35: Resolution instructions, published baselines and a single-call baseline. First row: accuracy (%, one-way strict) with no flags; other rows: paired difference from it in points. RAMDocs uses the multi-answer prompt (it has same-name questions with several answers); ConflictBank and its majority variant (three wrong-side passages, one correct) use the single-answer prompt and gpt-4o-mini. *Lab.*: labelled RAMDocs questions. 3 seeds, except †: gpt-4.1 seed 0 only, relative to the seed-0 no-flag accuracy in the first row (other gpt-4.1 cells are relative to its 3-seed no-flag accuracy, Table 33). *instr.*: the v2 reliability instruction. *(multi)*: the baseline's final step asks for every valid answer. *Single call*: one call that names suspect passages, then answers. Calls per question; *1+1*: one detector call and one generator call. **Bold**: significant under the paper's single scheme.

		Detector					Accuracy with				Flag minus	
Detector	Generator	Right	Wrong	Gen. acc.	Score	Gain	flag	verdict	hint	det. alone	hint	det. alone
gpt-4o-mini	gpt-6.1-terra	67.0	30.3	73.0	−4.7	**−11.7**	61.3	–	67.7	51.1	−6.3	**+10.2**
gpt-4o-mini	Claude Haiku	67.0	30.3	66.8	+1.9	−4.1	62.7	61.7	64.1	51.1	−1.4	**+11.6**
gpt-6.1-terra	gpt-6.1-terra	29.7	4.3	73.0	+2.7	+2.7	75.7	–	66.0	73.0	**+9.7**	+2.7
gpt-4o-mini	gpt-4.1	67.0	30.3	64.7	+4.0	−2.3	62.3	–	61.7	51.1	+0.7	**+11.2**
gpt-6.1-terra	Claude Haiku	29.7	4.3	66.8	+4.3	**+4.1**	70.9	68.2	59.9	73.0	**+11.0**	−2.1
gpt-6.1-terra	gpt-4.1	29.7	4.3	64.7	+4.7	**+5.0**	69.7	–	68.0	73.0	+1.7	−3.3
gpt-6-luna	gpt-6.1-terra	45.7	6.7	73.0	+5.3	+2.7	75.7	–	66.0	53.6	**+9.7**	**+22.1**
gpt-6-luna	Claude Haiku	45.7	6.7	66.8	+7.1	**+5.0**	71.8	69.7	61.4	53.6	**+10.3**	**+18.2**
gpt-6.1-terra	gpt-6-luna	29.7	4.3	53.6	+7.7	**+7.3**	60.9	60.6	66.7	73.0	−5.8	**−12.1**
gpt-6-luna	gpt-4.1	45.7	6.7	64.7	+7.7	**+6.3**	71.0	–	61.3	53.6	**+9.7**	**+17.4**
Claude Haiku	Claude Haiku	61.7	11.3	66.8	+9.8	**+5.7**	72.4	71.6	61.3	66.8	**+11.1**	**+5.7**
Claude Sonnet	Claude Haiku	65.0	9.7	66.8	+11.7	**+8.2**	75.0	73.8	60.4	75.7	**+14.6**	−0.7
gpt-6.1-terra	Qwen2.5-7B	31.0	4.0	50.3	+11.7	**+8.0**	58.3	55.0	62.3	73.0	−4.0	**−14.7**
gpt-6.1-terra	gpt-4o-mini	31.3	4.0	51.1	+12.8	**+9.6**	60.7	58.4	60.6	73.0	+0.1	**−12.3**
gpt-6-luna	gpt-6-luna	45.7	6.7	53.6	+13.8	**+11.1**	64.7	65.3	56.4	53.6	**+8.2**	**+11.1**
gpt-4o-mini	gpt-6-luna	67.0	30.3	53.6	+14.7	**+8.6**	62.1	61.9	56.1	51.1	+6.0	**+11.0**
gpt-4.1	gpt-6.1-terra	79.0	8.0	73.0	+15.7	**+7.3**	80.3	–	69.7	64.7	**+10.7**	**+15.7**
gpt-4o-mini	Qwen2.5-7B	67.0	30.3	50.3	+17.3	**+7.3**	57.7	54.7	53.3	51.1	+4.3	+6.6
gpt-4o-mini	gpt-4o-mini	67.0	30.3	51.1	+17.6	**+8.4**	59.6	57.1	52.4	51.1	+7.1	**+8.4**
gpt-6-luna	gpt-4o-mini	45.7	6.7	51.1	+19.1	**+13.7**	64.8	63.3	56.7	53.6	**+8.1**	**+11.2**
Claude Haiku	gpt-6-luna	61.7	11.3	53.6	+19.1	**+14.3**	67.9	68.7	63.0	66.8	+4.9	+1.1
gpt-6-luna	Qwen2.5-7B	48.0	6.7	50.3	+19.7	**+12.0**	62.3	59.7	58.0	53.6	+4.3	**+8.8**
gpt-4.1	Claude Haiku	79.0	8.0	66.8	+20.2	**+11.0**	77.8	76.0	63.7	64.7	**+14.1**	**+13.1**
gpt-4.1	gpt-4.1	79.0	8.0	64.7	+20.7	**+11.7**	76.3	–	63.3	64.7	**+13.0**	**+11.7**
Claude Haiku	gpt-4o-mini	61.7	11.3	51.1	+21.3	**+13.0**	64.1	62.7	57.9	66.8	+6.2	−2.7
Claude Sonnet	gpt-6-luna	65.0	9.7	53.6	+22.8	**+18.1**	71.7	72.9	67.8	75.7	+3.9	−4.0
Claude Sonnet	gpt-4o-mini	65.0	9.7	51.1	+28.1	**+15.9**	67.0	69.0	61.2	75.7	+5.8	**−8.7**
gpt-4.1	gpt-6-luna	79.0	8.0	53.6	+30.1	**+21.3**	74.9	76.8	61.2	64.7	**+13.7**	**+10.2**
gpt-4.1	gpt-4o-mini	79.0	8.0	51.1	+33.8	**+18.6**	69.7	67.6	57.9	64.7	**+11.8**	+5.0
gpt-4.1	Qwen2.5-7B	79.3	7.7	50.3	+36.7	**+16.0**	66.3	63.3	57.7	64.7	+8.7	+1.7

Table 36: Every detector–generator cell of H24 and H27 (ConflictBank, one-answer prompt, 300 conflict questions; %), sorted by the exploratory score. *Right*: the detector's hit rate, the share of questions on which it flags the true contradiction and names exactly its misinformation passage (computed on the detections that cell's generator saw; its direction precision, the same share among questions where it flags the true pair, pooled over its cells, is gpt-4o-mini 68.8%, gpt-4.1 81.8%, gpt-6-luna 82.2%, gpt-6.1-terra 82.6%, Claude Haiku 78.1% and Claude Sonnet 77.7%); *Wrong*: share on which it marks a correct passage as suspect; *Gen. acc.*: the generator's accuracy with no flags; *Score*: the mean over the 300 questions of [right direction and the generator errs] minus [a correct passage marked and the generator is right]; *Gain*: the LLM directional flag's change in points. *Verdict*: the detector's judgement passed as an answer key ("the answer X is false" for the answer its suspect passage supports; H30). *Hint*: the generator given the detector model's own plain answer, with no passage marked (H27); *det. alone*: the detector model's own plain accuracy. A detector's statistics can differ slightly between generators because concurrent runs made independent detector calls (up to 29 of 400 questions differ for one detector). Bold: significant under the paper's scheme (Gain: exploratory family; hint and det. alone columns: H27, Holm across cells).

Cell level (30 cells): accuracy with the flag, A	
$A \sim D + G$: coefficient of D (hit rate)	0.09 [0.03, 0.16]
coefficient of G	0.46 [0.36, 0.55]
$A \sim D$ + generator FE, permutation of D	$p = 0.29$
$A \sim P + R$ + gen. FE: precision P	1.03 [0.76, 1.20]; $p = 0.047$
recall R	0.16 [0.05, 0.26]; $p = 0.040$
Leave one detector out: R^2 for the gain	
G only	0.17
gap $D - G$ (registered)	-0.21
$P + R + G$	0.12
score (exploratory)	0.62
Question level, question/detector/generator FE (points)	
detector right on q	$+34.4$ [$+30.7$, $+38.3$]
detector marks a correct passage	-31.3 [-36.7, -26.5]

	H24 cells (observational)			H31 (randomised)	
Generator	right	wrong	wrong, abst.	right	wrong
Qwen2.5-7B	61	68	6	–	–
Claude Haiku	57	59	18	63	64
gpt-4.1	69	65	13	–	–
gpt-4o-mini	66	52	24	57	51
gpt-6-luna	75	57	16	72	77
gpt-6.1-terra	68	75	4	67	95

Table 37: H24 re-analysed (§15.4, §16.4; data already seen). D: hit rate (the detector flags the gold pair and names exactly its misinformation passage); P: direction precision (of those gold-pair flags, the share naming it); R: recall (the share of questions whose gold pair is flagged); G: the generator's no-flag accuracy. CIs: 2,000 question resamples shared across cells (300 with question fixed effects; the leave-one-detector-out range of the question-level coefficients is $+32.7$ to $+37.1$ and -32.8 to -26.8). p: exact permutation of the detectors' values across the six detectors; the design is unbalanced (the Claude detectors serve three generators), so detectors are only approximately exchangeable. Lower table, one definition of following: a right flag is followed when the flagged answer is correct (questions the generator gets wrong in every plain seed); a wrong flag is followed when the flagged answer contains the misinformation (questions it gets right in every plain seed); *abst.*: the flagged answer abstains instead. With no flag, a held-out plain seed changes outcome on at most 12% of such questions.

Policy	Setting	Neutral flag	Policy	Setting	Neutral flag
forced	CB forced, gpt-6.1-terra	−0.3	multi	RAMDocs, Claude Haiku	+1.2
forced	CB forced, gpt-4o-mini	−0.2	one	CB one-answer, gpt-4o-mini	−18.7
forced	CB forced, gpt-6-luna	+1.6	one	CB one-answer, gpt-6-luna	−9.4
forced	CB forced, Claude Haiku	+3.9	one	CB one-answer, gpt-4.1	−8.0
minimal	CB minimal, gpt-4o-mini	−18.1	one	CB majority, gpt-6.1-terra	−3.7
minimal	RAMDocs, gpt-4.1 (minimal)	−11.0	one	CB one-answer, Qwen2.5-7B	−3.3
minimal	RAMDocs, Qwen2.5-7B (minimal)	−5.8	one	CB one-answer, gpt-6.1-terra	−2.3
minimal	RAMDocs, gpt-4o-mini (minimal)	−5.3	one	RAGuard, gpt-6-luna	0.0
multi	CB multi-answer, gpt-4o-mini	−8.8	one	RAGuard, Claude Haiku	+0.3
multi	RAMDocs, gpt-4.1 (multi-answer)	−8.6	one	CB one-answer, Claude Haiku	+0.7
multi	CB multi-answer, Qwen2.5-7B	−8.3	one	CB majority, gpt-4o-mini	+1.1
multi	RAMDocs, gpt-4o-mini (multi-answer)	−6.7	one	RAGuard, gpt-6.1-terra	+2.4
multi	CB multi-answer, gpt-4.1	−6.3	one	CB one-answer, Claude Sonnet	+3.0
multi	RAMDocs, Qwen2.5-7B (multi-answer)	−5.8	one	RAGuard, gpt-4o-mini	+3.2
multi	RAMDocs, gpt-6-luna	−3.1	one	CB majority, Claude Haiku	+4.3
multi	RAMDocs, gpt-6.1-terra	−0.7	one	RAGuard, Claude Sonnet	+4.4
multi	RAMDocs, Claude Sonnet	0.0	one	CB majority, gpt-6-luna	+6.6
Multilevel model (random effects for generator, benchmark and setting; REML)					
pooled, all settings		−1.4 [−5.1, +2.3]; prediction interval [−12.8, +10.0]			
pooled, multi-answer prompt (10 settings)		−3.5 [−6.4, −0.6]			
pooled, minimal prompt (4 settings)		−12.1 [−23.2, −1.0]			
pooled, one answer prompt (16 settings)		−0.4 [−5.3, +4.5]			
pooled, forced prompt (4 settings)		+1.5 [−1.7, +4.8]			
Meta-regression (same model)					
intercept (OpenAI, one answer, generated)		−2.1 [−7.3, +3.0]			
several answers allowed		−3.7 [−8.2, +0.8]			
forced (no unknown)		+6.0 [+1.7, +10.3]			
Anthropic		+4.6 [+1.5, +7.7]			
Qwen		+0.6 [−3.9, +5.1]			
natural misinformation (RAGuard)		+2.2 [−7.2, +11.6]			

Table 38: The 34 settings on misinformation benchmarks (RAMDocs, ConflictBank and its one-answer, minimal, forced-choice and majority variants, RAGuard) in which the oracle neutral flag was run; WikiContradict (genuine conflicts) and CONFLICTS are not pooled (conflict questions, change in points), grouped by answer policy. The pooled estimate was registered as exploratory (§13.4) and the multilevel model, the policy split and the moderators in §16.4; they use data already seen. The prediction interval describes this design space (our generators, benchmarks and prompts), not deployments in general. *Natural misinformation* is RAGuard only; RAMDocs' misinformation is synthetic. The moderator *several answers allowed* codes the multi-answer and minimal prompts together, against one-answer prompts; the policy rows above separate them. Settings share questions, so the model is fitted with their sampling covariance, estimated by a joint question bootstrap (§17.1: each replicate resamples each benchmark's questions once for every setting that uses them). Treating the settings as independent gives [−5.5, +2.2]; conditional on these settings (question sampling only) the pooled effect is [−2.2, −0.6].

	Neutral flag, utility change at misinformation cost c				Dir. flag	Neutral flag: plain answer → flagged answer (null)			
Setting	$c=0$	$c=1$	$c=2$	$c=4$	$c=2$	mis.→abst.	mis.→corr.	corr.→abst.	corr.→mis.
RAMDocs, gpt-4o-mini (multi-answer)	**−6.7**	**−20.9**	**−35.0**	**−63.3**	**+79.1**	1 (0)	2 (2)	0 (0)	29 (3)
RAMDocs, gpt-6-luna	**−3.1**	−7.7	−12.2	−21.3	**+160.9**	0 (0)	1 (3)	0 (0)	22 (16)
RAMDocs, Claude Haiku	+1.2	+9.1	**+17.0**	**+32.9**	**+125.2**	0 (0)	9 (4)	0 (0)	5 (9)
CB multi-answer, gpt-4o-mini	**−8.8**	**−19.2**	**−29.7**	**−50.6**	**+91.0**	0 (0)	4 (3)	1 (0)	55 (9)
CB one-answer, gpt-4o-mini	**−18.7**	−5.3	+8.0	**+34.7**	**+83.7**	45 (2)	7 (3)	37 (1)	5 (2)
CB one-answer, gpt-6-luna	**−9.4**	−2.6	+4.3	+18.1	**+85.1**	27 (10)	12 (13)	20 (6)	5 (6)
CB one-answer, gpt-6.1-terra	−2.3	0.0	+2.3	+7.0	**+51.7**	2 (–)	26 (–)	4 (–)	5 (–)
CB one-answer, Claude Haiku	+0.7	+0.6	+0.4	+0.2	**+66.9**	0 (0)	25 (16)	0 (0)	8 (5)
CB one-answer, Claude Sonnet	+3.0	+6.7	+10.3	+17.7	**+45.7**	2 (–)	34 (–)	0 (–)	3 (–)
CB forced, gpt-4o-mini	−0.2	−0.6	−0.9	−1.6	**+74.0**	1 (0)	19 (4)	0 (0)	11 (2)
CB majority, gpt-4o-mini	+1.1	**+24.6**	**+48.0**	**+94.9**	**+118.2**	26 (1)	11 (2)	22 (1)	8 (5)
RAGuard, gpt-4o-mini	+3.2	+6.6	+10.0	+16.8	**+34.6**	12 (3)	7 (3)	8 (6)	4 (2)
RAGuard, gpt-6-luna	0.0	−1.1	−2.3	−4.5	**+29.3**	10 (9)	6 (9)	8 (9)	7 (8)
RAGuard, Claude Haiku	+0.3	−1.8	−4.0	−8.3	**+26.5**	7 (6)	4 (4)	11 (9)	6 (3)

Table 39: Abstention-aware view of the oracle flags (conflict questions; §15.4 and §16.4, re-analyses of data already seen). Utility per answer: $+1$ correct, $-c$ if it contains the misinformation, 0 for abstaining or any other wrong answer; $c = 0$ is strict accuracy. Bold: significant in one Holm family per c over all 34 settings of Table 38 (a selection is shown). Right: of the question–seed pairs whose plain answer contained the misinformation (or was correct), the share whose flagged answer abstained, was correct or contained the misinformation; in brackets the same share between two plain runs with different seeds, the churn with no flag.

Setting	Contrast	Matcher (points)	Judge (points)	Judge − matcher
RAMDocs, gpt-4o-mini	neutral flag vs plain	−4.2 [−9.2, 0.0]	−2.5 [−9.2, +4.2]	+1.7 [−3.3, +6.7]
ConflictBank, gpt-4o-mini	neutral flag vs plain	−20.0 [−27.5, −12.5]	−21.7 [−29.2, −14.2]	−1.7 [−4.2, 0.0]
ConflictBank, Claude Haiku	neutral flag vs plain	+3.3 [−3.3, +10.0]	+6.7 [+0.8, +13.3]	+3.3 [0.0, +7.5]
RAMDocs, gpt-4o-mini	oracle directional flag vs answer key	−0.8 [−5.8, +4.2]	+0.8 [−8.3, +10.0]	+1.7 [−6.7, +10.0]
ConflictBank, gpt-4o-mini	oracle directional flag vs answer key	+10.8 [+4.2, +17.5]	+12.5 [+6.7, +19.2]	+1.7 [−1.7, +5.0]
ConflictBank, gpt-6-luna	oracle directional flag vs answer key	−1.7 [−5.8, +2.5]	0.0 [−3.3, +3.3]	+1.7 [−1.7, +5.0]
ConflictBank, gpt-4o-mini	LLM directional flag vs verdict key	+3.3 [−2.5, +9.2]	+6.7 [+0.8, +12.5]	+3.3 [0.0, +7.5]
ConflictBank, gpt-6-luna	LLM directional flag vs verdict key	−0.8 [−3.3, +1.7]	0.0 [−2.5, +2.5]	+0.8 [−1.7, +4.2]
ConflictBank, gpt-4o-mini	randomised directional flag vs plain	+4.2 [−5.8, +15.0]	+6.7 [−3.3, +16.7]	+2.5 [0.0, +5.8]
ConflictBank, Claude Haiku	randomised directional flag vs plain	−7.5 [−18.3, +3.3]	−6.7 [−17.5, +4.2]	+0.8 [−1.7, +3.3]

Condition	Outputs	Matcher (%)	Judge (%)	Judge only (%)	Matcher only (%)
answer key	360	67.8	77.8	10.3	0.3
directional flag (LLM)	240	59.6	62.5	3.8	0.8
directional flag (oracle)	360	70.6	82.2	11.7	0.0
neutral flag (oracle)	360	36.9	43.3	6.9	0.6
no flag	600	48.8	53.3	5.2	0.7
randomised direction (H31)	240	54.6	59.6	5.0	0.0
verdict key (LLM)	240	58.3	59.2	2.1	1.2

Table 40: Semantic re-score (§17.3; exploratory). An outside-family judge (gemini-3.5-flash-lite, temperature 0) reads the question, the reference and known wrong answers and the response, and classifies the final answer; judge-strict means it gives the reference answer, asserts no wrong one and does not abstain. Top: each contrast on the same 120 conflict questions (seed 0) under the matcher and the judge, and their difference, with question-bootstrap 95% CIs. Bottom: per condition, the share scored correct by each and the shares where only one does. The judge is an LLM given the answer keys: it checks the matcher's semantics, not the benchmark labels.

Generator	Oracle flag	Single-entity	Multi-entity	Gap
Qwen2.5-7B	neutral	−12.2 [−24.4, 0.0]	−3.1 [−8.2, +2.0]	−9.1
Qwen2.5-7B	directional	+9.8 [+2.4, +19.5]	+1.0 [−4.1, +6.1]	+8.7
Qwen2.5-7B, minimal	neutral	−4.9 [−17.1, +7.3]	−6.1 [−11.2, −2.0]	+1.2
Qwen2.5-7B, minimal	directional	+17.1 [0.0, +34.1]	−3.1 [−8.2, +2.0]	+20.1
claude-haiku-5-5	neutral	+5.7 [−0.8, +13.8]	−0.7 [−5.4, +3.7]	+6.4
claude-haiku-5-5	directional	+43.9 [+29.3, +58.5]	+21.8 [+13.3, +30.6]	+22.1
claude-sonnet-5-5	neutral	−4.9 [−14.6, +4.9]	+2.0 [−3.1, +7.1]	−6.9
claude-sonnet-5-5	directional	+17.1 [+4.9, +29.3]	+13.3 [+5.1, +21.4]	+3.8
gpt-4.1	neutral	−5.7 [−13.0, 0.0]	−9.9 [−15.3, −4.4]	+4.2
gpt-4.1	directional	+13.0 [+4.1, +23.6]	+20.4 [+12.9, +28.6]	−7.4
gpt-4.1, minimal	neutral	−13.0 [−26.0, 0.0]	−10.2 [−17.7, −2.7]	−2.8
gpt-4.1, minimal	directional	+23.6 [+9.8, +37.4]	+8.2 [+0.3, +16.0]	+15.4
gpt-4o-mini	neutral	−3.3 [−9.8, +2.4]	−8.2 [−13.6, −3.7]	+4.9
gpt-4o-mini	directional	+38.2 [+24.4, +53.7]	+3.1 [−3.7, +9.9]	+35.2
gpt-4o-mini, minimal	neutral	−3.3 [−11.4, +4.9]	−6.1 [−11.2, −2.0]	+2.9
gpt-4o-mini, minimal	directional	+31.7 [+18.7, +45.5]	−6.1 [−12.6, −0.3]	+37.8
gpt-6-luna	neutral	−2.4 [−5.7, 0.0]	−3.4 [−6.1, −1.4]	+1.0
gpt-6-luna	directional	+51.2 [+36.6, +65.0]	+34.0 [+24.8, +43.2]	+17.2
gpt-6.1-terra	neutral	0.0 [−7.3, +7.3]	−1.0 [−3.1, 0.0]	+1.0
gpt-6.1-terra	directional	+46.3 [+31.7, +61.0]	+36.7 [+26.5, +45.9]	+9.6

Table 41: RAMDocs hold-out, oracle flag minus no flag on conflict questions, in points, split by whether the correct passages give one answer (41 *single-entity* questions, where no oracle pair can join passages about different entities) or several (98), with question-bootstrap 95% CIs (§18.2; exploratory).

				Flag names the false side			Flag marks a correct passage			Rest	Gain (points)		
Generator	Detector	Prec.	Rec.	Share	Acc.	Gain	Share	Acc.	Gain	contrib.	Marginal	Question	Observed
gpt-4o-mini	Claude Haiku	78	79	62	58	+28.8	11	42	−30.4	−1.3	+16.1	+12.3	+13.0
gpt-4o-mini	Claude Sonnet	78	84	65	51	+32.8	10	40	−20.7	−3.4	+15.8	+16.1	+15.9
gpt-4o-mini	gpt-4.1	82	97	79	55	+28.6	8	25	−22.2	−2.2	+18.3	+19.2	+18.6
gpt-4o-mini	gpt-4o-mini	69	97	67	60	+23.1	30	30	−23.1	0.0	+10.5	+10.4	+8.4
gpt-4o-mini	gpt-6-luna	82	56	46	50	+36.7	7	57	−46.7	0.0	+18.5	+11.0	+13.7
gpt-4o-mini	gpt-6.1-terra	83	38	31	51	+36.9	4	61	−36.1	−0.6	+19.2	+7.4	+9.6
gpt-6-luna	Claude Haiku	78	79	62	61	+31.5	11	45	−34.3	−1.2	+16.3	+13.5	+14.3
gpt-6-luna	Claude Sonnet	78	84	65	59	+34.5	10	41	−35.6	−0.9	+16.0	+16.2	+18.1
gpt-6-luna	gpt-4.1	82	97	79	60	+31.1	8	21	−20.8	−1.6	+19.2	+21.6	+21.3
gpt-6-luna	gpt-4o-mini	69	97	67	63	+27.0	30	33	−31.5	0.0	+9.1	+10.0	+8.6
gpt-6-luna	gpt-6-luna	82	56	46	64	+29.2	7	40	−35.0	+0.1	+19.4	+9.8	+11.1
gpt-6-luna	gpt-6.1-terra	82	36	30	66	+30.3	4	56	−38.5	0.0	+19.7	+5.4	+7.3
Claude Haiku	Claude Haiku	78	79	62	79	+14.8	11	28	−22.5	−0.9	+8.0	+6.1	+5.7
Claude Haiku	Claude Sonnet	78	84	65	74	+18.1	10	53	−37.9	+0.1	+7.8	+7.3	+8.2
Claude Haiku	gpt-4.1	82	97	79	73	+18.0	8	17	−16.7	−1.9	+10.7	+12.7	+11.0
Claude Haiku	gpt-4o-mini	69	97	67	78	+12.6	30	43	−41.4	0.0	+1.3	+1.1	−4.1
Claude Haiku	gpt-6-luna	82	56	46	78	+14.8	7	45	−15.0	−0.8	+10.9	+4.4	+5.0
Claude Haiku	gpt-6.1-terra	82	36	30	76	+18.0	4	67	−28.2	0.0	+11.2	+2.7	+4.1
gpt-6.1-terra	gpt-4.1	82	97	79	74	+16.9	8	62	−62.5	−1.0	+2.6	+9.0	+7.3
gpt-6.1-terra	gpt-4o-mini	69	97	67	79	+10.4	30	62	−61.5	0.0	−8.0	−8.4	−11.7
gpt-6.1-terra	gpt-6-luna	82	56	46	77	+13.1	7	80	−45.0	−0.3	+2.9	+2.1	+2.7
gpt-6.1-terra	gpt-6.1-terra	82	36	30	80	+14.6	4	77	−30.8	−0.3	+3.2	+0.8	+2.7
Qwen2.5-7B	gpt-4.1	82	97	79	50	+27.3	8	43	−39.1	−2.7	–	+23.4	+16.0
Qwen2.5-7B	gpt-4o-mini	69	97	67	54	+25.9	30	44	−33.0	0.0	–	+10.2	+7.3
Qwen2.5-7B	gpt-6-luna	83	58	48	50	+30.6	7	65	−25.0	−1.0	–	+12.4	+12.0
Qwen2.5-7B	gpt-6.1-terra	83	37	31	54	+25.8	4	67	−16.7	+0.7	–	+7.3	+8.0
gpt-4.1	gpt-4.1	82	97	79	72	+19.8	8	21	−20.8	−2.3	–	+13.2	+11.7
gpt-4.1	gpt-4o-mini	69	97	67	77	+13.9	30	38	−38.5	0.0	–	+1.7	−2.3
gpt-4.1	gpt-6-luna	82	56	46	75	+18.2	7	55	−35.0	+0.3	–	+4.7	+6.3
gpt-4.1	gpt-6.1-terra	82	36	30	74	+20.2	4	69	−23.1	0.0	–	+2.8	+5.0

Table 42: The crossed cells (ConflictBank, one-answer prompt, conflict questions) split by what the flag does on each question (exploratory, §17.1). *Prec.*, *Rec.*: direction precision and recall of the gold pair (%). *Share*: share of the cell's questions; *Acc.*: the generator's no-flag accuracy on them (%); *Gain*: the flag's mean effect on them (points). *Rest*: the remaining questions' contribution to the gain. *Marginal*: the break-even calculation, $pR + (1-p)W$ with H31's R and W at the cell's precision p, which assumes every question is flagged on its true pair and that flagged questions are typical; *Question*: the follow-rate account from each question's own flag and no-flag answer; both are retrospective. Generators without an H31 run have no marginal prediction; their question-level prediction uses mean follow rates.

		Conflict questions					Clean	
Generator	Detector	Right	Wrong	Acc.	Pred.	Obs.	Pred.	Obs.
Claude Haiku	Qwen2.5-7B	59	32	59.6	+3.2	+6.2	−16.4	−8.3
Claude Haiku	Claude Haiku	69	7	59.6	+12.7	+14.6	0.0	0.0
Claude Haiku	gpt-4o-mini	72	27	59.6	+7.9	+7.8	−5.8	−2.3
Claude Haiku	gpt-6-luna	43	9	59.6	+4.2	+7.4	0.0	0.0
Claude Haiku	gpt-6.1-terra	33	2	59.6	+5.8	+8.1	0.0	0.0
gpt-4o-mini	Qwen2.5-7B	59	32	47.4	+9.7	+9.2	−12.2	−5.3
gpt-4o-mini	Claude Haiku	69	7	47.4	+17.7	+16.0	0.0	0.0
gpt-4o-mini	gpt-4o-mini	72	27	47.4	+14.2	+11.4	−4.1	−3.7
gpt-4o-mini	gpt-6-luna	43	9	47.4	+8.0	+9.1	0.0	0.0
gpt-4o-mini	gpt-6.1-terra	33	2	47.4	+9.2	+10.2	0.0	0.0
gpt-6-luna	Qwen2.5-7B	59	32	52.7	+8.1	+6.9	−19.6	−2.7
gpt-6-luna	Claude Haiku	69	7	52.7	+18.4	+17.3	0.0	0.0
gpt-6-luna	gpt-4o-mini	72	27	52.7	+13.7	+11.6	−6.4	−3.0
gpt-6-luna	gpt-6-luna	43	9	52.7	+7.4	+6.1	0.0	0.0
gpt-6-luna	gpt-6.1-terra	33	2	52.7	+8.2	+8.8	0.0	0.0
gpt-6.1-terra	Qwen2.5-7B	59	32	71.7	−10.1	−8.3	−25.6	0.0
gpt-6.1-terra	Claude Haiku	69	7	71.7	+7.4	+7.0	0.0	0.0
gpt-6.1-terra	gpt-4o-mini	72	27	71.7	−3.1	−4.0	−8.5	−2.0
gpt-6.1-terra	gpt-6-luna	43	9	71.7	−1.7	−0.7	0.0	0.0
gpt-6.1-terra	gpt-6.1-terra	33	2	71.7	+3.0	+2.3	0.0	0.0

Table 43: Prospective test on ConflictBank-fresh (§19.1; 300 conflict and 100 clean questions no earlier run used). *Right*/*Wrong*: % of questions on which the detector names exactly the false side of a gold pair / marks a correct passage. *Acc.*: no-flag accuracy. *Pred.*: the frozen prediction, committed before any flagged answer existed; *Obs.*: observed directional-flag effect, in points.

			Effect against no flag (points)			
Generator	No flag	Closed-book	Oracle neutral	Oracle directional	LLM directional	Pred.
gpt-4o-mini	60.5	76.1	−2.4 [−6.0, +1.1]	+5.5 [+1.1, +10.1]	−2.7 [−6.8, +1.3]	−8.2
gpt-6-luna	68.0	91.0	−0.1 [−2.2, +2.0]	+11.7 [+7.7, +15.9]	−8.6 [−14.9, −2.8]	−19.4
Claude Haiku	69.6	91.2	−0.4 [−2.7, +2.0]	+13.8 [+9.7, +18.5]	+1.1 [−3.0, +5.2]	−16.6

Table 44: HealthContradict (§19.3): yes/no health questions with one real web page per side, accuracy in % and effects in points with topic-cluster bootstrap 95% CIs. *Pred.*: the frozen follow-rate prediction for the LLM directional flag (gpt-4o-mini detector), committed before any flagged answer existed.